STAT 437 Final Project_Life Exp

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library

```
# install.packages("party")
# install.packages("rpart")
# install.packages("nlme")
```

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(lattice)
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(geepack)
```

```
## Warning: package 'geepack' was built under R version 4.3.3
```

```r
library(rpart)
```

```
## Warning: package 'rpart' was built under R version 4.3.3
```

```r
library(nlme)
```

```
## Warning: package 'nlme' was built under R version 4.3.3
```

```
##
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```r
library(caret)
library(randomForest)
```

```
## randomForest 4.7-1.1
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
library(randomForestExplainer)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
```

```r
library(mgcv)
```

```
## This is mgcv 1.9-0. For overview type 'help("mgcv-package")'.
```

```r
library(lme4)
```

```
## Loading required package: Matrix
```

```
## Warning in check_dep_version(): ABI version mismatch:
## lme4 was built with Matrix ABI version 1
## Current Matrix ABI version is 0
## Please re-install lme4 from source or restore original 'Matrix' package
```

```
##
## Attaching package: 'lme4'
```

```
## The following object is masked from 'package:nlme':
##
##     lmList
```

1. Data loading and cleaning

```
# reading the dataset - the fixed one
life = read.csv("Life-Expectancy-Data-Updated.csv")

#frequency table for year and country
year_frequency = table(life$Year)
country_frequency = table(life$Country)

year_frequency
```

```
##
## 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015
##  179  179  179  179  179  179  179  179  179  179  179  179  179  179  179  179
```

```
#country_frequency
```

- This is a balanced data - every subject(country) gets the same number of observations

1.1 checking multicollinearity

```
life_no_country <- life %>% select(-Country, -Economy_status_Developing) #get rid of economy status dev
    # plot(life_2015_mod)
model_multi <- lm(Life_expectancy ~ ., data = life_no_country)
vif_values <- vif(model_multi)
print(vif_values)
```

```
##                                 GVIF Df GVIF^(1/(2*Df))
## Region                     53.762164  8        1.282788
## Year                        1.140513  1        1.067948
## Infant_deaths              48.182565  1        6.941366
## Under_five_deaths          50.042479  1        7.074071
## Adult_mortality             8.109518  1        2.847722
## Alcohol_consumption         3.385143  1        1.839876
## Hepatitis_B                 2.743537  1        1.656363
```

```
## Measles                       1.661728  1        1.289080
## BMI                           3.901192  1        1.975144
## Polio                        12.177095  1        3.489569
## Diphtheria                   13.228137  1        3.637051
## Incidents_HIV                 3.030843  1        1.740932
## GDP_per_capita                2.647151  1        1.627007
## Population_mln                1.213073  1        1.101396
## Thinness_ten_nineteen_years   8.994848  1        2.999141
## Thinness_five_nine_years      9.213216  1        3.035328
## Schooling                     5.528179  1        2.351208
## Economy_status_Developed      7.282037  1        2.698525
```

There might be multicolinearity issue with some variables:

- Infant_deaths
- Under_five_deaths
- Thinness_ten_nineteen_years
- Thinness_five_nine_years
- Polio and Diphtheria - measurement of if they got the vaccine

Take out infant_deaths

They are super highly correlated

```
cor(life_no_country$Infant_deaths, life_no_country$Under_five_deaths)
```

```
## [1] 0.9856513
```

```
cor(life_no_country$Polio, life_no_country$Diphtheria)
```

```
## [1] 0.9531779
```

```
cor(life_no_country$Thinness_ten_nineteen_years, life_no_country$Thinness_five_nine_years)
```

```
## [1] 0.9387571
```

```
life_mod = life %>% select(- Infant_deaths, -Under_five_deaths, -Adult_mortality, - Thinness_five_nine_y
```

1.2 modifying the region Africa

```
countries <- c("Algeria", "Egypt", "Libya", "Morocco", "Tunisia")

# Modify the "Region" column
life_mod <- life_mod%>%
  mutate(Region = ifelse(Country %in% countries, "Rest of Africa", Region))

life_mod <- life_mod %>%
  mutate(Region = ifelse(Region == "Africa", "Sub-Saharan Africa", Region))

table(life_mod$Region)
```

```
##
##                          Asia Central America and Caribbean
##                           432                             304
##                 European Union                     Middle East
##                           432                             224
##                  North America                         Oceania
##                             48                             176
##                 Rest of Africa                Rest of Europe
##                             64                             240
##                  South America            Sub-Saharan Africa
##                            192                             752
```

2. EDA

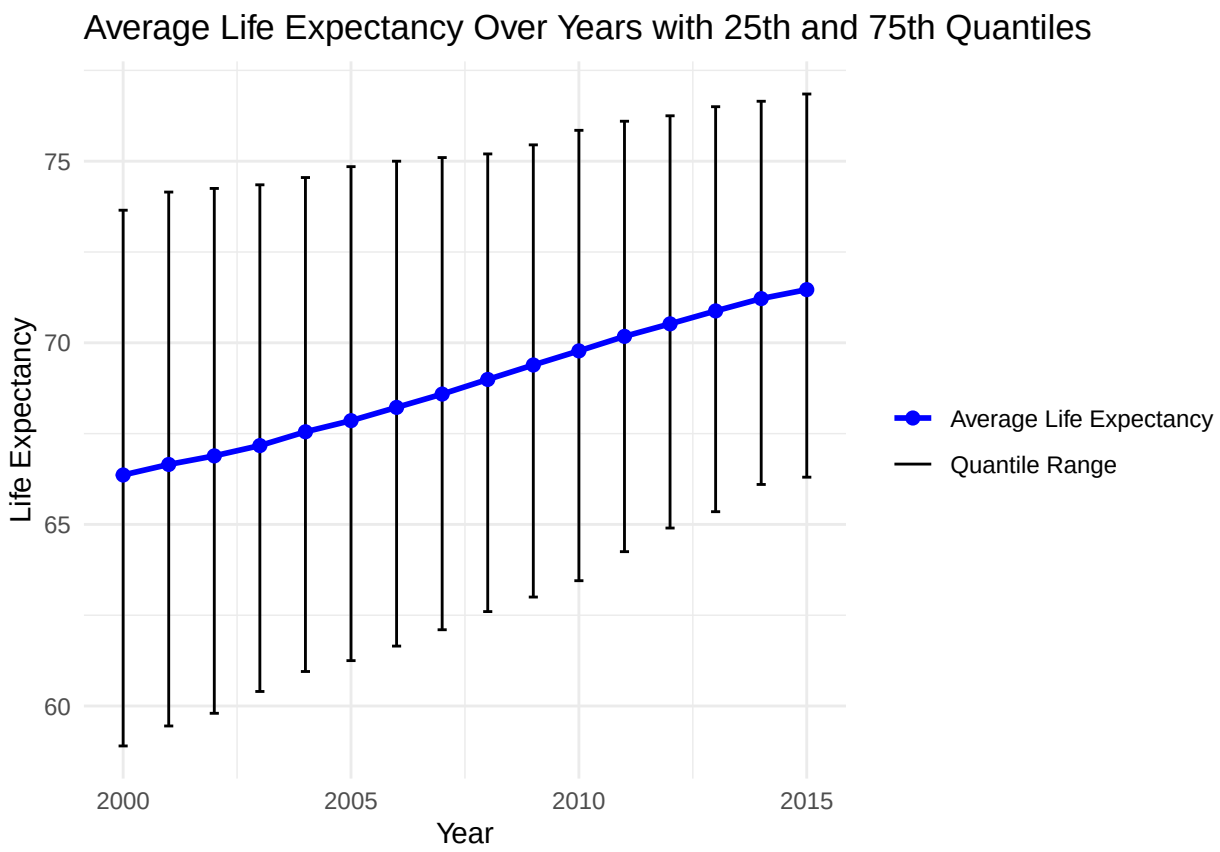
2.1 Response variable

Global average life expectancy over years

```
summary_data <- life_mod %>%
  group_by(Year) %>%
  summarise(
    Avg_Life_Expectancy = mean(Life_expectancy),
    Q25_Life_Expectancy = quantile(Life_expectancy, 0.25),
    Q75_Life_Expectancy = quantile(Life_expectancy, 0.75)
  )

ggplot(summary_data, aes(x = Year)) +
  geom_line(aes(y = Avg_Life_Expectancy, color = "Average Life Expectancy"), size = 1) +  # Average lif
  geom_errorbar(aes(ymin = Q25_Life_Expectancy, ymax = Q75_Life_Expectancy, color = "Quantile Range"),
                width = 0.2) +  # Error bars
  geom_point(aes(y = Avg_Life_Expectancy, color = "Average Life Expectancy"), size = 2) +
  scale_color_manual(values = c("Average Life Expectancy" = "blue", "Quantile Range" = "black")) +
  labs(title = "Average Life Expectancy Over Years with 25th and 75th Quantiles",
       x = "Year",
       y = "Life Expectancy",
       color = "") +
  theme_minimal()
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

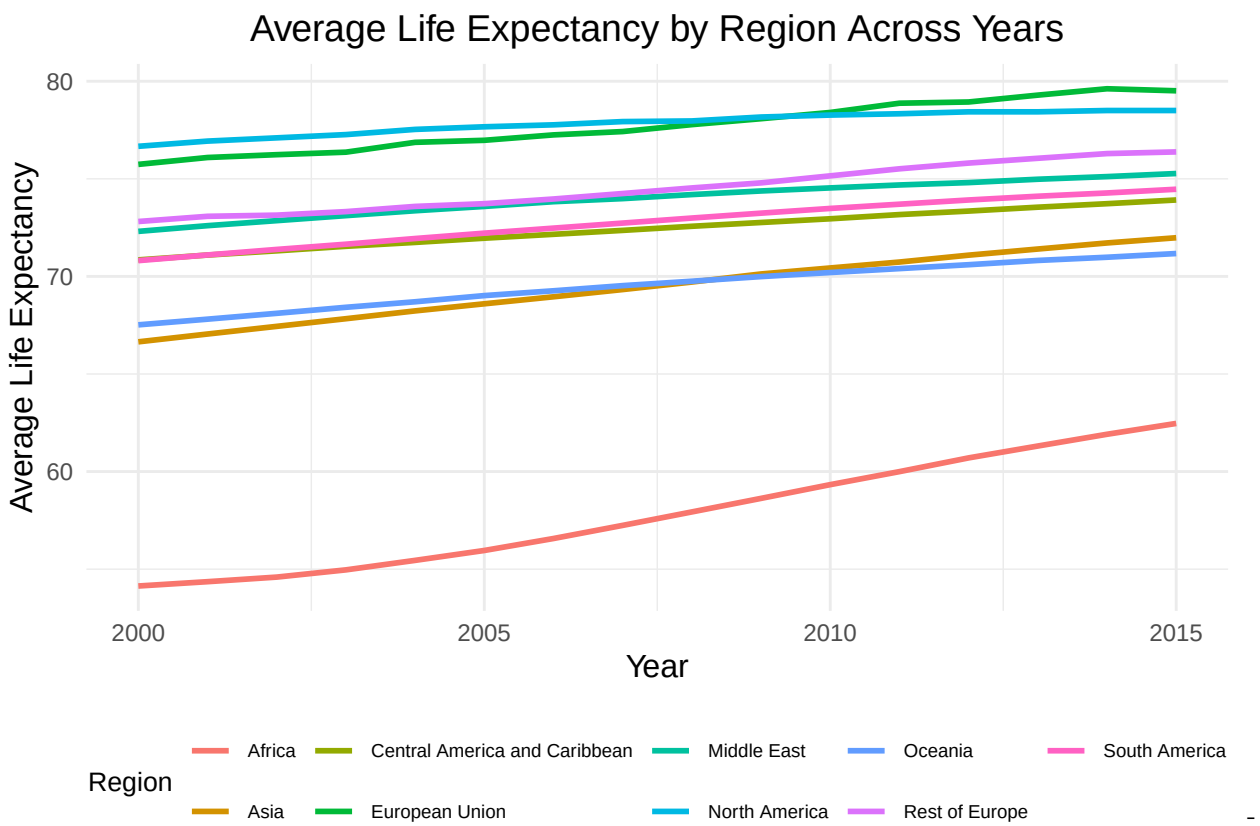


2.2 Some univaraite analysis of important variables

Region

```
avg_life_expectancy <- life %>%
  group_by(Year, Region) %>%
  summarise(Average_Life_Expectancy = mean(Life_expectancy, na.rm = TRUE), .groups = "drop")

ggplot(avg_life_expectancy, aes(x = Year, y = Average_Life_Expectancy, color = Region, group = Region))
  geom_line(size = 1) +
  labs(title = "Average Life Expectancy by Region Across Years",
       x = "Year",
       y = "Average Life Expectancy",
       color = "Region") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(size = 7),  # Adjust the size as needed
        legend.title = element_text(size = 10),
        plot.title = element_text(size = 14, hjust = 0.5),
        axis.title = element_text(size = 12))
```



We can clearly see that different region has different life expectancy, with sub-saharan africa region being a far outlier. We also observe that the rate of increase for SSA region is a lot faster than other countries in the recent years.

```
life_2015 = life_mod[life_mod$Year == "2015", ]

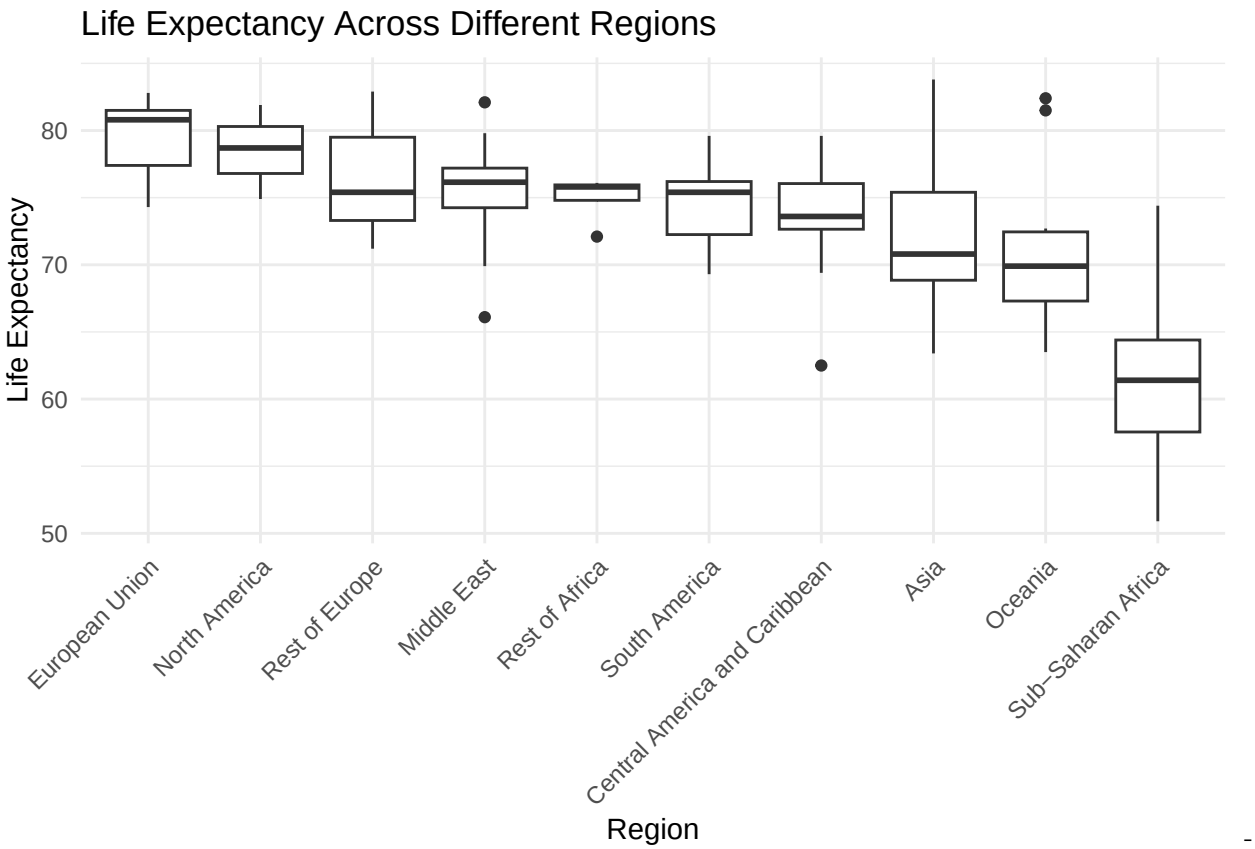
life_2015 <- life_2015 %>%
  mutate(Economy_status_Developed = as.factor(Economy_status_Developed))

life_2015_mod <- life_2015 %>% select(-Year, -Country, -Economy_status_Developing)
```

```
region_order <- life_2015_mod %>%
  group_by(Region) %>%
  summarise(mean_life_expectancy = mean(Life_expectancy, na.rm = TRUE)) %>%
  arrange(desc(mean_life_expectancy))  # Use `arrange()` for descending order

# Convert Region to a factor ordered by mean life expectancy
life_2015_mod <- life_2015_mod %>%
  mutate(Region = factor(Region, levels = region_order$Region))

# Create the boxplot
ggplot(life_2015_mod, aes(x = Region, y = Life_expectancy)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Life Expectancy Across Different Regions",
       x = "Region",
       y = "Life Expectancy") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

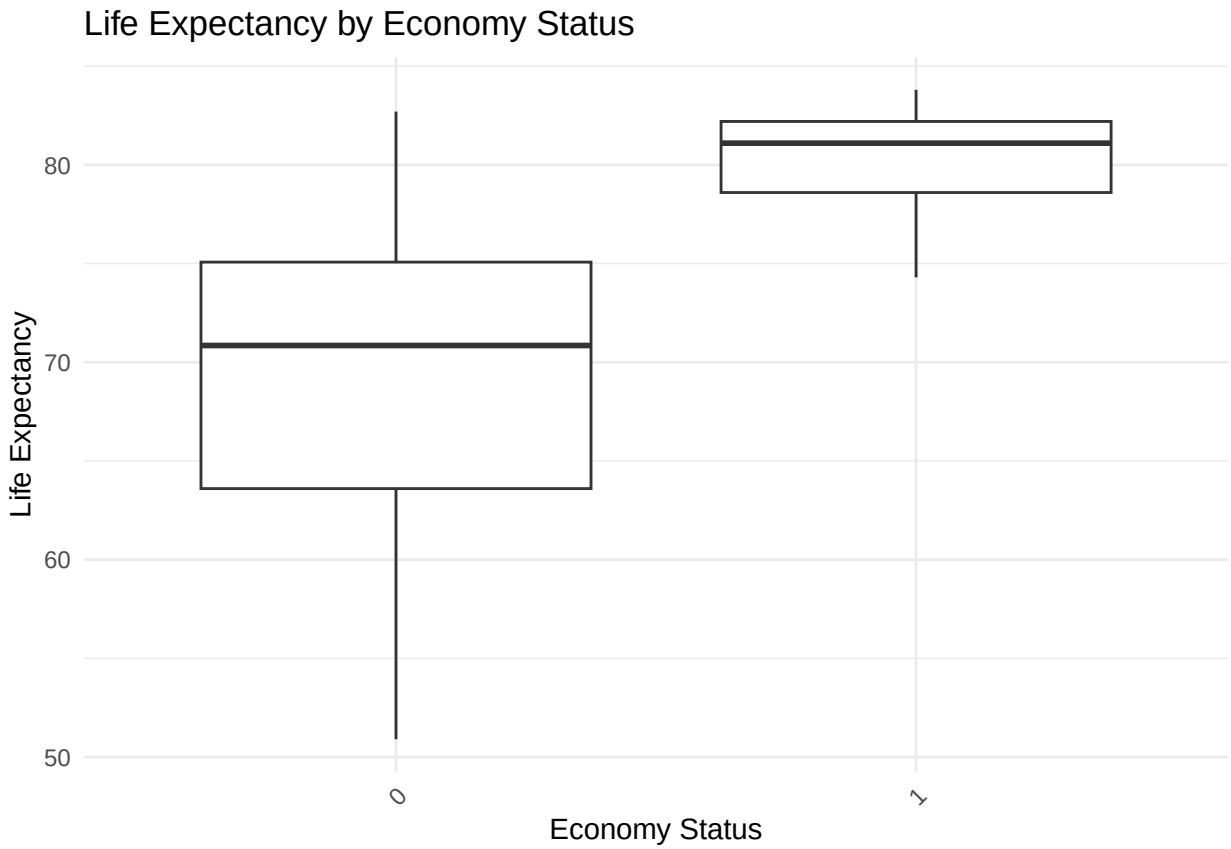


Even if between region there are big differences, per region boxplot is still quite wide, meaning the countries within a region is not very homogeneous.

- This suggest we should keep country as ID.

By development status

```
# Create the box pl
ggplot(life_2015_mod, aes(x = Economy_status_Developed, y = Life_expectancy)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Life Expectancy by Economy Status",
       x = "Economy Status",
       y = "Life Expectancy") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Ensure labels are readable
```



Choice of predictors to include in the model:

```
life_final = life_mod %>% select(- Economy_status_Developed)
colnames(life_final)
```

```
##  [1] "Country"                    "Region"
##  [3] "Year"                       "Alcohol_consumption"
##  [5] "Hepatitis_B"                "Measles"
##  [7] "BMI"                        "Polio"
##  [9] "Incidents_HIV"              "GDP_per_capita"
## [11] "Population_mln"             "Thinness_ten_nineteen_years"
## [13] "Schooling"                  "Economy_status_Developing"
## [15] "Life_expectancy"
```

- Country is the ID, the rest are the explanatory variables

```
head(life_final)
```

```
##      Country                      Region Year Alcohol_consumption Hepatitis_B
## 1    Turkiye                  Middle East 2015                1.32          97
## 2      Spain                European Union 2015               10.35          97
## 3      India                         Asia 2007                1.57          60
## 4     Guyana                 South America 2006                5.68          93
## 5     Israel                  Middle East 2012                2.89          97
## 6 Costa Rica Central America and Caribbean 2006                4.19          88
##   Measles  BMI Polio Incidents_HIV GDP_per_capita Population_mln
```

```
## 1      65 27.8    97          0.08          11006          78.53
## 2      94 26.0    97          0.09          25742          46.44
## 3      35 21.2    67          0.13           1076        1183.21
## 4      74 25.3    92          0.79           4146           0.75
## 5      89 27.0    94          0.08          33995           7.91
## 6      86 26.4    89          0.16           9110           4.35
##   Thinness_ten_nineteen_years Schooling Economy_status_Developing
## 1                         4.9       7.8                         1
## 2                         0.6       9.7                         0
## 3                        27.1       5.0                         1
## 4                         5.7       7.9                         1
## 5                         1.2      12.8                         0
## 6                         2.0       7.9                         1
##   Life_expectancy
## 1            76.5
## 2            82.8
## 3            65.4
## 4            67.0
## 5            81.7
## 6            78.2
```

3. Fitting Model - Entire Dataset

3.0 Pre-LME

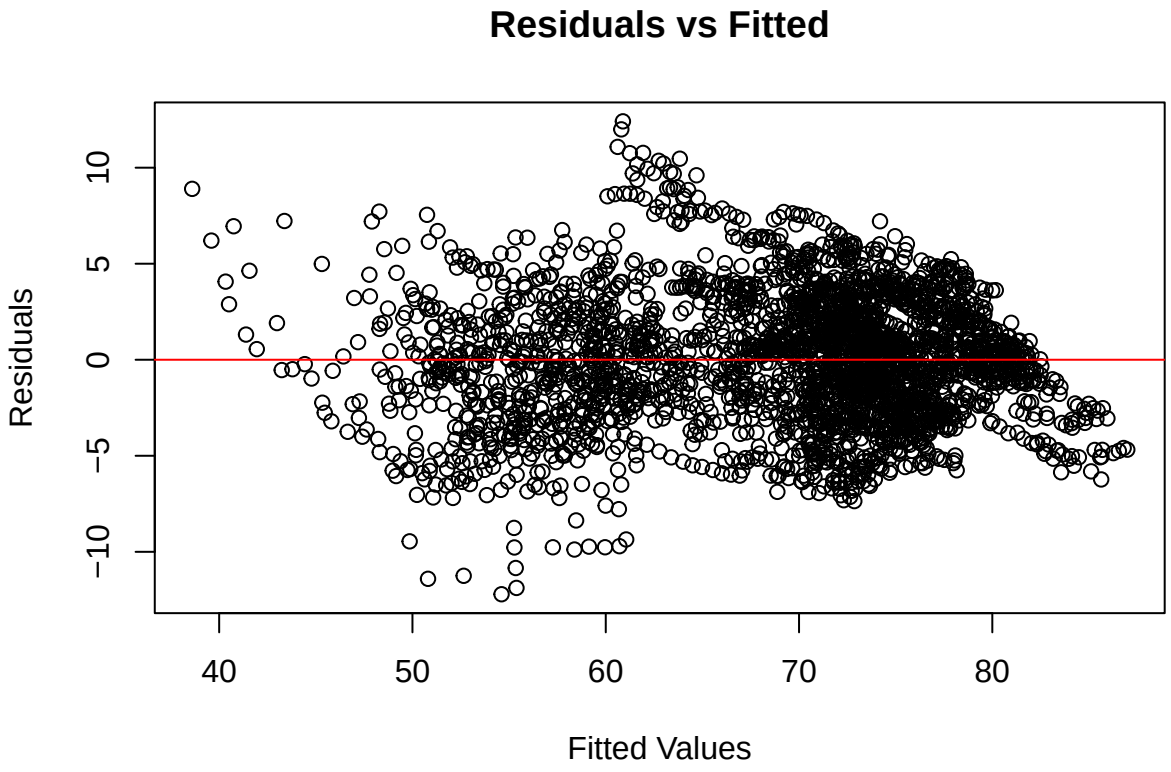
Most regular linear regression model

```
lin_mod = lm(Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + BMI + Polio
                                  Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_nineteen_years
                                  Schooling + Economy_status_Developing,
             data = life_final)
summary(lin_mod)
```

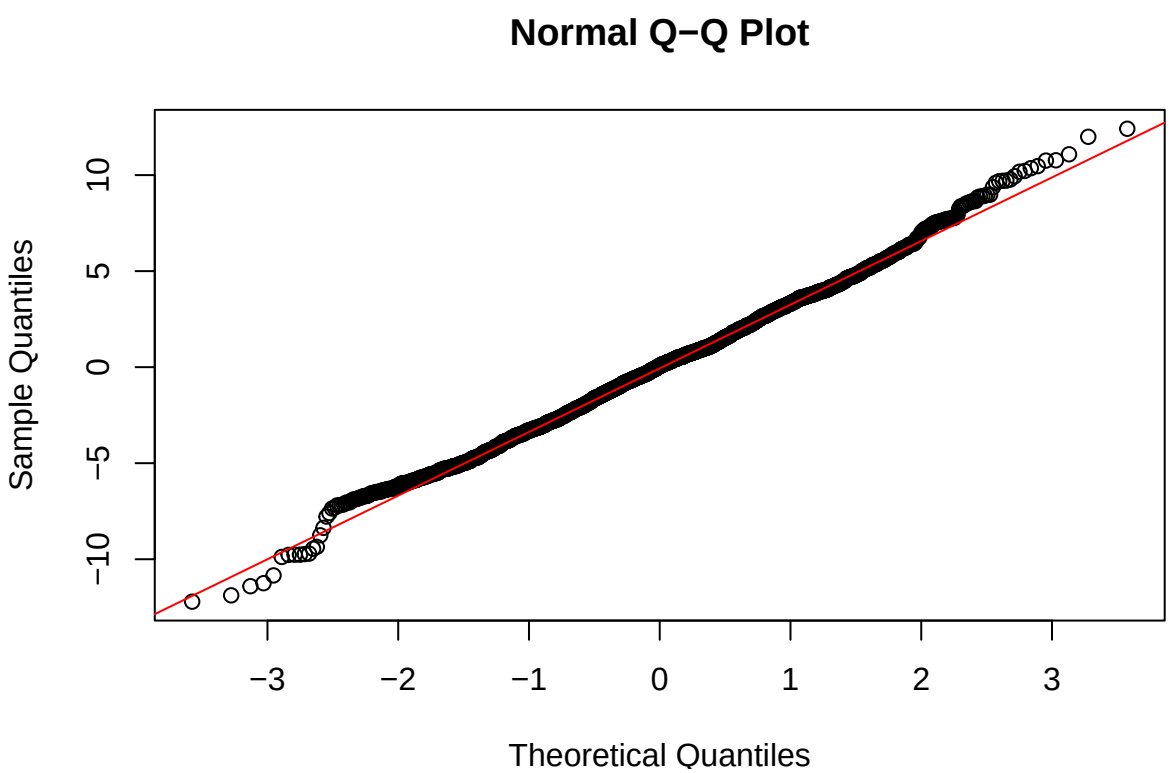
```
##
## Call:
## lm(formula = Life_expectancy ~ Year + Region + Alcohol_consumption +
##     Hepatitis_B + Measles + BMI + Polio + Incidents_HIV + GDP_per_capita +
##     Population_mln + Thinness_ten_nineteen_years + Schooling +
##     Economy_status_Developing, data = life_final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2091  -2.2969   0.0582   2.1732  12.4120
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      -2.652e+02  2.851e+01  -9.299  < 2e-16 ***
## Year                              1.558e-01  1.434e-02  10.867  < 2e-16 ***
## RegionCentral America and Caribbean  2.296e+00  3.121e-01   7.357 2.45e-13 ***
## RegionEuropean Union             -2.453e-01  4.444e-01  -0.552  0.58098
## RegionMiddle East                 1.550e+00  3.371e-01   4.597 4.48e-06 ***
```

```
## RegionNorth America                -9.887e-02  5.888e-01  -0.168  0.86665
## RegionOceania                      -1.052e+00  3.767e-01  -2.792  0.00527 **
## RegionRest of Africa                2.615e+00  4.788e-01   5.462 5.12e-08 ***
## RegionRest of Europe                5.966e-01  3.283e-01   1.817  0.06930 .
## RegionSouth America                 2.862e+00  3.446e-01   8.304  < 2e-16 ***
## RegionSub-Saharan Africa           -5.637e+00  2.730e-01 -20.646  < 2e-16 ***
## Alcohol_consumption                -1.110e-01  2.844e-02  -3.903 9.74e-05 ***
## Hepatitis_B                        -3.471e-03  5.902e-03  -0.588  0.55653
## Measles                             7.060e-03  4.269e-03   1.654  0.09829 .
## BMI                                 3.677e-01  5.476e-02   6.716 2.25e-11 ***
## Polio                               1.470e-01  7.080e-03  20.767  < 2e-16 ***
## Incidents_HIV                      -1.018e+00  3.214e-02 -31.679  < 2e-16 ***
## GDP_per_capita                      9.356e-05  5.633e-06  16.608  < 2e-16 ***
## Population_mln                      2.355e-03  4.968e-04   4.741 2.23e-06 ***
## Thinness_ten_nineteen_years         1.668e-02  2.123e-02   0.786  0.43201
## Schooling                           4.499e-01  4.305e-02  10.449  < 2e-16 ***
## Economy_status_Developing          -4.121e+00  4.064e-01 -10.141  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.314 on 2842 degrees of freedom
## Multiple R-squared:  0.8768, Adjusted R-squared:  0.8759
## F-statistic: 963.1 on 21 and 2842 DF,  p-value: < 2.2e-16
```

```r
# residual vs. fitted
plot(lin_mod$fitted.values, residuals(lin_mod),
     main = "Residuals vs Fitted",
     xlab = "Fitted Values",
     ylab = "Residuals")
abline(h = 0, col = "red")
```




```
# qq Plot
qqnorm(residuals(lin_mod))
qqline(residuals(lin_mod), col = "red")
```



- From the fitted vs. residual plot, seems like there's some fanning pattern - the residuals show more variance for

countries (* year) with lower life expectancy. This suggest that maybe we should allow different countries to have different intercept/slope to capture those variances.

GEE model

```
gee <- geeglm(Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + BMI + Poli
                                  Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_ninet
                                  Schooling + Economy_status_Developing,
                    id = Country,
                    data = life_final,
                    corstr = "exchangeable",
                    family = gaussian)
summary(gee)
```

3.1 Fit LME model

random intercept model

```
library(nlme)
lme_int <- lme(fixed = Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + I
                                  Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_ninet
                                  Schooling + Economy_status_Developing,
               random = ~ 1 | Country,
               data = life_final, na.action = na.omit)
summary(lme_int)
```

```
## Linear mixed-effects model fit by REML
##   Data: life_final
##        AIC     BIC    logLik
##   9415.956 9558.81 -4683.978
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept) Residual
## StdDev:    4.230153 1.033807
##
## Fixed effects:  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B +      Measles +
##                                         Value Std.Error   DF   t-value p-value
## (Intercept)                          -517.5989 21.039556 2674 -24.60123  0.0000
## Year                                    0.2950  0.011612 2674  25.40239  0.0000
## RegionCentral America and Caribbean     3.6697  1.334126  168   2.75061  0.0066
## RegionEuropean Union                   -1.3077  1.965739  168  -0.66524  0.5068
## RegionMiddle East                       4.4215  1.476299  168   2.99498  0.0032
## RegionNorth America                     2.0926  2.805876  168   0.74579  0.4568
## RegionOceania                          -0.8141  1.609603  168  -0.50577  0.6137
## RegionRest of Africa                    3.8956  2.299460  168   1.69414  0.0921
## RegionRest of Europe                    3.0517  1.467239  168   2.07992  0.0391
## RegionSouth America                     3.8007  1.515677  168   2.50763  0.0131
## RegionSub-Saharan Africa               -8.8460  1.069721  168  -8.26945  0.0000
## Alcohol_consumption                     0.0526  0.024878 2674   2.11374  0.0346
```

```
## Hepatitis_B                              0.0025  0.002779 2674   0.91234  0.3617
## Measles                                 -0.0129  0.002877 2674  -4.48901  0.0000
## BMI                                      0.0335  0.112677 2674   0.29721  0.7663
## Polio                                    0.0595  0.003358 2674  17.72030  0.0000
## Incidents_HIV                           -1.1226  0.034498 2674 -32.54128  0.0000
## GDP_per_capita                           0.0000  0.000012 2674   0.09386  0.9252
## Population_mln                           0.0029  0.001869 2674   1.54149  0.1233
## Thinness_ten_nineteen_years             -0.0192  0.011573 2674  -1.65829  0.0974
## Schooling                               -0.1524  0.056980 2674  -2.67426  0.0075
## Economy_status_Developing               -9.8183  1.665372  168  -5.89558  0.0000
##  Correlation:
##                                    (Intr) Year   RgCAaC RgnErU RgnMdE RgnNrA
## Year                               -0.994
## RegionCentral America and Caribbean -0.177  0.166
## RegionEuropean Union               -0.147  0.079  0.276
## RegionMiddle East                  -0.221  0.209  0.426  0.301
## RegionNorth America                -0.165  0.134  0.200  0.420  0.217
## RegionOceania                      -0.194  0.174  0.392  0.378  0.377  0.240
## RegionRest of Africa               -0.073  0.067  0.261  0.153  0.242  0.111
## RegionRest of Europe               -0.208  0.173  0.407  0.472  0.388  0.284
## RegionSouth America                -0.150  0.139  0.403  0.238  0.365  0.174
## RegionSub-Saharan Africa            0.130 -0.154  0.495  0.297  0.430  0.189
## Alcohol_consumption                -0.136  0.136 -0.042 -0.060  0.063  0.002
## Hepatitis_B                         0.081 -0.079 -0.011 -0.008 -0.005  0.011
## Measles                             0.039 -0.041 -0.019 -0.015 -0.024 -0.012
## BMI                                 0.700 -0.745 -0.223 -0.131 -0.273 -0.161
## Polio                               0.050 -0.047  0.001  0.009  0.002  0.005
## Incidents_HIV                      -0.218  0.210  0.005 -0.005  0.030  0.013
## GDP_per_capita                      0.182 -0.197 -0.039  0.104 -0.114 -0.018
## Population_mln                      0.111 -0.118  0.176  0.118  0.129 -0.014
## Thinness_ten_nineteen_years         0.016 -0.021  0.055  0.032  0.020  0.021
## Schooling                           0.480 -0.451 -0.001 -0.007  0.019 -0.018
## Economy_status_Developing           0.028 -0.097 -0.054  0.769  0.011  0.334
##                                    RgnOcn RgnRoA RgnRoE RgnStA RgS-SA Alchl_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa                0.221
## RegionRest of Europe                0.389  0.223
## RegionSouth America                 0.337  0.224  0.352
## RegionSub-Saharan Africa            0.399  0.289  0.420  0.429
## Alcohol_consumption                 0.019  0.028 -0.053 -0.041 -0.053
## Hepatitis_B                        -0.005 -0.005  0.002 -0.010 -0.004  0.043
## Measles                            -0.002 -0.026 -0.026 -0.001  0.010 -0.051
## BMI                                -0.256 -0.115 -0.179 -0.181  0.058 -0.126
## Polio                               0.020 -0.003 -0.006  0.004  0.021 -0.071
## Incidents_HIV                       0.013  0.008  0.011  0.004 -0.106  0.034
## GDP_per_capita                     -0.006 -0.017 -0.021 -0.024  0.041 -0.114
## Population_mln                      0.141  0.088  0.143  0.122  0.212 -0.057
## Thinness_ten_nineteen_years         0.046  0.015  0.050  0.055  0.047 -0.077
## Schooling                           0.018  0.039 -0.080 -0.011  0.148 -0.021
```

```
## Economy_status_Developing            0.140 -0.022  0.219 -0.046 -0.025  0.023
##                                    Hptt_B Measls BMI    Polio  In_HIV GDP_p_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles                           -0.193
## BMI                                0.027  0.006
## Polio                             -0.354 -0.058 -0.077
## Incidents_HIV                     -0.001 -0.060 -0.099  0.100
## GDP_per_capita                    -0.023 -0.008  0.114  0.051 -0.101
## Population_mln                     0.005 -0.025  0.058 -0.073 -0.035  0.041
## Thinness_ten_nineteen_years        0.019 -0.001  0.025 -0.040 -0.142 -0.008
## Schooling                         -0.041 -0.035 -0.055  0.075 -0.014 -0.113
## Economy_status_Developing          -0.012 -0.001  0.007  0.029 -0.026  0.271
##                                    Ppltn_ Thn___ Schlng
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI
## Polio
## Incidents_HIV
## GDP_per_capita
## Population_mln
## Thinness_ten_nineteen_years        0.041
## Schooling                          0.010  0.056
## Economy_status_Developing          0.012 -0.020  0.088
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -6.11484506 -0.42295445 -0.02410951  0.40086603  8.93794278
##
## Number of Observations: 2864
## Number of Groups: 179
```

randome intercept + slope model

```
lme_slope = lme(fixed = Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles +
                                  Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_ninet
                                  Schooling + Economy_status_Developing,
                random = ~ Year | Country,
                data = life_final,
                na.action = na.omit)
summary(lme_slope)
```

```
## Linear mixed-effects model fit by REML
##   Data: life_final
##        AIC      BIC    logLik
##   9419.956 9574.714 -4683.978
##
## Random effects:
##  Formula: ~Year | Country
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev       Corr
## (Intercept) 4.230154e+00 (Intr)
## Year        3.592580e-07 0.001
## Residual    1.033807e+00
##
## Fixed effects:  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B +      Measles +
##                                         Value Std.Error   DF   t-value p-value
## (Intercept)                          -517.5989 21.039558 2674 -24.60123  0.0000
## Year                                    0.2950  0.011612 2674  25.40239  0.0000
## RegionCentral America and Caribbean     3.6697  1.334127  168   2.75061  0.0066
## RegionEuropean Union                   -1.3077  1.965740  168  -0.66524  0.5068
## RegionMiddle East                       4.4215  1.476299  168   2.99498  0.0032
## RegionNorth America                     2.0926  2.805877  168   0.74579  0.4568
## RegionOceania                          -0.8141  1.609604  168  -0.50577  0.6137
## RegionRest of Africa                    3.8956  2.299461  168   1.69414  0.0921
## RegionRest of Europe                    3.0517  1.467240  168   2.07992  0.0391
## RegionSouth America                     3.8008  1.515677  168   2.50762  0.0131
## RegionSub-Saharan Africa               -8.8460  1.069722  168  -8.26944  0.0000
## Alcohol_consumption                     0.0526  0.024878 2674   2.11374  0.0346
## Hepatitis_B                             0.0025  0.002779 2674   0.91234  0.3617
## Measles                                -0.0129  0.002877 2674  -4.48901  0.0000
## BMI                                     0.0335  0.112677 2674   0.29721  0.7663
## Polio                                   0.0595  0.003358 2674  17.72030  0.0000
## Incidents_HIV                          -1.1226  0.034498 2674 -32.54128  0.0000
## GDP_per_capita                          0.0000  0.000012 2674   0.09386  0.9252
## Population_mln                          0.0029  0.001869 2674   1.54149  0.1233
## Thinness_ten_nineteen_years            -0.0192  0.011573 2674  -1.65829  0.0974
## Schooling                              -0.1524  0.056980 2674  -2.67427  0.0075
## Economy_status_Developing              -9.8183  1.665372  168  -5.89558  0.0000
##  Correlation:
##                                     (Intr) Year   RgCAaC RgnErU RgnMdE RgnNrA
## Year                                -0.994
## RegionCentral America and Caribbean -0.177  0.166
## RegionEuropean Union                -0.147  0.079  0.276
## RegionMiddle East                   -0.221  0.209  0.426  0.301
```

```
## RegionNorth America                -0.165  0.134  0.200  0.420  0.217
## RegionOceania                      -0.194  0.174  0.392  0.378  0.377  0.240
## RegionRest of Africa               -0.073  0.067  0.261  0.153  0.242  0.111
## RegionRest of Europe               -0.208  0.173  0.407  0.472  0.388  0.284
## RegionSouth America                -0.150  0.139  0.403  0.238  0.365  0.174
## RegionSub-Saharan Africa            0.130 -0.154  0.495  0.297  0.430  0.189
## Alcohol_consumption                -0.136  0.136 -0.042 -0.060  0.063  0.002
## Hepatitis_B                         0.081 -0.079 -0.011 -0.008 -0.005  0.011
## Measles                             0.039 -0.041 -0.019 -0.015 -0.024 -0.012
## BMI                                 0.700 -0.745 -0.223 -0.131 -0.273 -0.161
## Polio                               0.050 -0.047  0.001  0.009  0.002  0.005
## Incidents_HIV                      -0.218  0.210  0.005 -0.005  0.030  0.013
## GDP_per_capita                      0.182 -0.197 -0.039  0.104 -0.114 -0.018
## Population_mln                      0.111 -0.118  0.176  0.118  0.129 -0.014
## Thinness_ten_nineteen_years         0.016 -0.021  0.055  0.032  0.020  0.021
## Schooling                           0.480 -0.451 -0.001 -0.007  0.019 -0.018
## Economy_status_Developing           0.028 -0.097 -0.054  0.769  0.011  0.334
##                                    RgnOcn RgnRoA RgnRoE RgnStA RgS-SA Alchl_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa                0.221
## RegionRest of Europe                0.389  0.223
## RegionSouth America                 0.337  0.224  0.352
## RegionSub-Saharan Africa            0.399  0.289  0.420  0.429
## Alcohol_consumption                 0.019  0.028 -0.053 -0.041 -0.053
## Hepatitis_B                        -0.005 -0.005  0.002 -0.010 -0.004  0.043
## Measles                            -0.002 -0.026 -0.026 -0.001  0.010 -0.051
## BMI                                -0.256 -0.115 -0.179 -0.181  0.058 -0.126
## Polio                               0.020 -0.003 -0.006  0.004  0.021 -0.071
## Incidents_HIV                       0.013  0.008  0.011  0.004 -0.106  0.034
## GDP_per_capita                     -0.006 -0.017 -0.021 -0.024  0.041 -0.114
## Population_mln                      0.141  0.088  0.143  0.122  0.212 -0.057
## Thinness_ten_nineteen_years         0.046  0.015  0.050  0.055  0.047 -0.077
## Schooling                           0.018  0.039 -0.080 -0.011  0.148 -0.021
## Economy_status_Developing           0.140 -0.022  0.219 -0.046 -0.025  0.023
##                                    Hptt_B Measls BMI    Polio  In_HIV GDP_p_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles                            -0.193
## BMI                                 0.027  0.006
```

```
## Polio                                -0.354 -0.058 -0.077
## Incidents_HIV                        -0.001 -0.060 -0.099  0.100
## GDP_per_capita                       -0.023 -0.008  0.114  0.051 -0.101
## Population_mln                        0.005 -0.025  0.058 -0.073 -0.035  0.041
## Thinness_ten_nineteen_years           0.019 -0.001  0.025 -0.040 -0.142 -0.008
## Schooling                            -0.041 -0.035 -0.055  0.075 -0.014 -0.113
## Economy_status_Developing            -0.012 -0.001  0.007  0.029 -0.026  0.271
##                                      Ppltn_ Thn___ Schlng
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI
## Polio
## Incidents_HIV
## GDP_per_capita
## Population_mln
## Thinness_ten_nineteen_years           0.041
## Schooling                            0.010  0.056
## Economy_status_Developing            0.012 -0.020  0.088
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -6.11484524 -0.42295448 -0.02410952  0.40086603  8.93794308
##
## Number of Observations: 2864
## Number of Groups: 179
```

Comparing

```
anova(lme_int,lme_slope)
```

```
##           Model df      AIC      BIC    logLik   Test      L.Ratio p-value
## lme_int       1 24 9415.956 9558.810 -4683.978
## lme_slope     2 26 9419.956 9574.714 -4683.978 1 vs 2 1.790959e-06       1
```

- Doing LRT test: it seems that the random slope didn't do that much - it is likely because the slope seems similar between different countries (they all show an upward trend).

```
AIC(lin_mod)
```

```
## [1] 15014.02
```

```r
# QIC(gee_model)
AIC(lme_int)
```

```
## [1] 9415.956
```

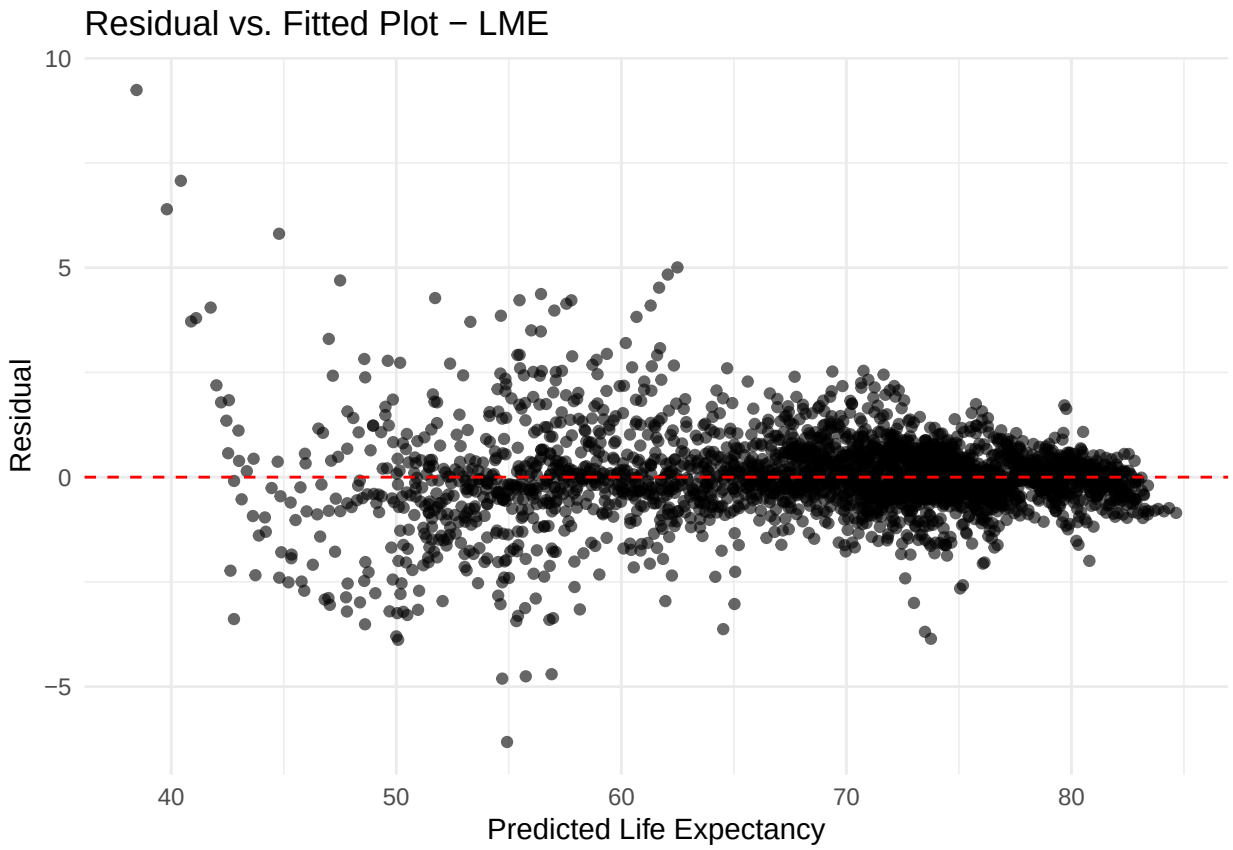
- The QIC for GEE model very high, and the AIC for LME model much smaller than lin mod. The best model so far is the LME model with the random intercept, it will be used to compare with MERT and MERF.

```r
fitted_vals <- fitted(lme_int)
residual_vals <- residuals(lme_int)

plot_data <- data.frame(
  Fitted = fitted_vals,
  Residual = residual_vals
)

library(ggplot2)

ggplot(plot_data, aes(x = Fitted, y = Residual)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(
    title = "Residual vs. Fitted Plot - LME",
    x = "Predicted Life Expectancy",
    y = "Residual"
  ) +
  theme_minimal()
```



3.2 MERT model

```
MERT <- function(data, response, fixed_effects, cluster, max_iter = 100, tol = 1e-4) {
  library(nlme)
  library(rpart)

  data <- data.frame(data)
  data$random_effect <- 0
  prev_logLik <- -Inf
  GLL_history <- numeric(max_iter)

  for (iter in 1:max_iter) {
    cat("Iteration:", iter, "\n")

    # subtract the random effect previous estimated
    data$adjusted_response <- data[[response]] - data$random_effect

    # fit the tree model to the adjusted response
    tree_formula <- as.formula(paste("adjusted_response ~", paste(fixed_effects, collapse = " + ")))
    tree_model <- rpart(tree_formula, data = data, method = "anova")
    # make prediction
    data$fixed_pred <- predict(tree_model, newdata = data)

    # take out the fix part of the effect and leave the randome to fit LME
    data$adjusted_y <- data[[response]] - data$fixed_pred
```

```
  # fit LME
  lme_model <- tryCatch({
    lme(adjusted_y ~ 1,
        random = as.formula(paste("~ 1 |", cluster)),
        data = data,
        na.action = na.omit,
        control = lmeControl(opt = "optim"))
  }, error = function(e) {
    cat("LME model failed to converge at iteration", iter, "\n")
    return(NULL)
  })

  if (is.null(lme_model)) break

  # Update randome effect from new LME model
  re <- ranef(lme_model)
  re_vector <- re[as.character(data[[cluster]]), "(Intercept)"]
  data$random_effect <- re_vector

  #gll formula
  sigma2 <- lme_model$sigma^2
  tau2 <- as.numeric(VarCorr(lme_model)[1, "Variance"])
  cluster_ids <- unique(data[[cluster]])
  GLL <- 0
  for (cid in cluster_ids) {
    subset <- data[data[[cluster]] == cid, ]
    y_i <- subset[[response]]
    f_Xi <- subset$fixed_pred
    b_i <- unique(subset$random_effect)
    Z_i <- matrix(1, nrow = length(y_i), ncol = 1)
    R_i <- sigma2 * diag(length(y_i))
    D <- matrix(tau2, 1, 1)

    resid <- y_i - f_Xi - Z_i %*% b_i
    term1 <- t(resid) %*% solve(R_i) %*% resid
    term2 <- t(b_i) %*% solve(D) %*% b_i
    term3 <- log(det(D))
    term4 <- log(det(R_i))

    GLL <- GLL + term1 + term2 + term3 + term4
  }
  GLL_history[iter] <- GLL
  cat("  GLL at iteration", iter, ":", GLL, "\n")

  # convergence and stopping
  if (iter > 1 && abs(GLL_history[iter] - GLL_history[iter - 1]) < tol) {
    cat("  Convergence reached at iteration", iter, "\n")
    break
  }
}

return(list(
  tree_model = tree_model,
```

```
    lme_model = lme_model,
    final_data = data,
    GLL = GLL_history[1:iter]
  ))
}
```

```
# fixed effect variables
fixed_vars <- c("Year", "Region", "Alcohol_consumption", "Hepatitis_B", "Measles", "BMI",
                "Polio", "Incidents_HIV", "GDP_per_capita", "Population_mln",
                "Thinness_ten_nineteen_years", "Schooling", "Economy_status_Developing")

#on full dataset
mert_result <- MERT(
  data = life_final,
  response = "Life_expectancy",
  fixed_effects = fixed_vars,
  cluster = "Country",
  max_iter = 100
)
```

```
## Iteration: 1
##   GLL at iteration 1 : 7054.883
## Iteration: 2
##   GLL at iteration 2 : 6858.956
## Iteration: 3
##   GLL at iteration 3 : 6426.721
## Iteration: 4
##   GLL at iteration 4 : 6361.74
## Iteration: 5
##   GLL at iteration 5 : 6285.379
## Iteration: 6
##   GLL at iteration 6 : 6239.036
## Iteration: 7
##   GLL at iteration 7 : 6213.071
## Iteration: 8
##   GLL at iteration 8 : 6171.969
## Iteration: 9
##   GLL at iteration 9 : 6151.635
## Iteration: 10
##   GLL at iteration 10 : 6134.933
## Iteration: 11
##   GLL at iteration 11 : 6120.899
## Iteration: 12
##   GLL at iteration 12 : 6068.015
## Iteration: 13
##   GLL at iteration 13 : 6058.843
## Iteration: 14
##   GLL at iteration 14 : 6051.254
## Iteration: 15
##   GLL at iteration 15 : 6044.888
## Iteration: 16
##   GLL at iteration 16 : 6146.291
## Iteration: 17
```

```
##   GLL at iteration 17 : 6125.47
## Iteration: 18
##   GLL at iteration 18 : 6108.939
## Iteration: 19
##   GLL at iteration 19 : 6092.518
## Iteration: 20
##   GLL at iteration 20 : 6072.603
## Iteration: 21
##   GLL at iteration 21 : 6050.749
## Iteration: 22
##   GLL at iteration 22 : 6047.312
## Iteration: 23
##   GLL at iteration 23 : 6044.34
## Iteration: 24
##   GLL at iteration 24 : 6041.938
## Iteration: 25
##   GLL at iteration 25 : 6039.996
## Iteration: 26
##   GLL at iteration 26 : 6038.425
## Iteration: 27
##   GLL at iteration 27 : 6037.154
## Iteration: 28
##   GLL at iteration 28 : 6036.125
## Iteration: 29
##   GLL at iteration 29 : 6035.292
## Iteration: 30
##   GLL at iteration 30 : 6034.618
## Iteration: 31
##   GLL at iteration 31 : 6034.071
## Iteration: 32
##   GLL at iteration 32 : 6033.627
## Iteration: 33
##   GLL at iteration 33 : 6033.268
## Iteration: 34
##   GLL at iteration 34 : 6032.976
## Iteration: 35
##   GLL at iteration 35 : 6032.739
## Iteration: 36
##   GLL at iteration 36 : 6032.546
## Iteration: 37
##   GLL at iteration 37 : 6032.389
## Iteration: 38
##   GLL at iteration 38 : 6032.261
## Iteration: 39
##   GLL at iteration 39 : 6032.157
## Iteration: 40
##   GLL at iteration 40 : 6032.072
## Iteration: 41
##   GLL at iteration 41 : 6032.003
## Iteration: 42
##   GLL at iteration 42 : 6031.946
## Iteration: 43
##   GLL at iteration 43 : 6031.899
## Iteration: 44
```

```
##   GLL at iteration 44 : 6031.861
## Iteration: 45
##   GLL at iteration 45 : 6031.83
## Iteration: 46
##   GLL at iteration 46 : 6031.804
## Iteration: 47
##   GLL at iteration 47 : 6031.783
## Iteration: 48
##   GLL at iteration 48 : 6031.765
## Iteration: 49
##   GLL at iteration 49 : 6031.75
## Iteration: 50
##   GLL at iteration 50 : 6031.738
## Iteration: 51
##   GLL at iteration 51 : 6031.728
## Iteration: 52
##   GLL at iteration 52 : 6031.72
## Iteration: 53
##   GLL at iteration 53 : 6031.713
## Iteration: 54
##   GLL at iteration 54 : 6031.707
## Iteration: 55
##   GLL at iteration 55 : 6031.702
## Iteration: 56
##   GLL at iteration 56 : 6031.698
## Iteration: 57
##   GLL at iteration 57 : 6031.694
## Iteration: 58
##   GLL at iteration 58 : 6031.691
## Iteration: 59
##   GLL at iteration 59 : 6031.689
## Iteration: 60
##   GLL at iteration 60 : 6031.687
## Iteration: 61
##   GLL at iteration 61 : 6031.685
## Iteration: 62
##   GLL at iteration 62 : 6031.683
## Iteration: 63
##   GLL at iteration 63 : 6031.682
## Iteration: 64
##   GLL at iteration 64 : 6031.681
## Iteration: 65
##   GLL at iteration 65 : 6031.68
## Iteration: 66
##   GLL at iteration 66 : 6031.679
## Iteration: 67
##   GLL at iteration 67 : 6031.678
## Iteration: 68
##   GLL at iteration 68 : 6031.678
## Iteration: 69
##   GLL at iteration 69 : 6031.677
## Iteration: 70
##   GLL at iteration 70 : 6031.676
## Iteration: 71
```

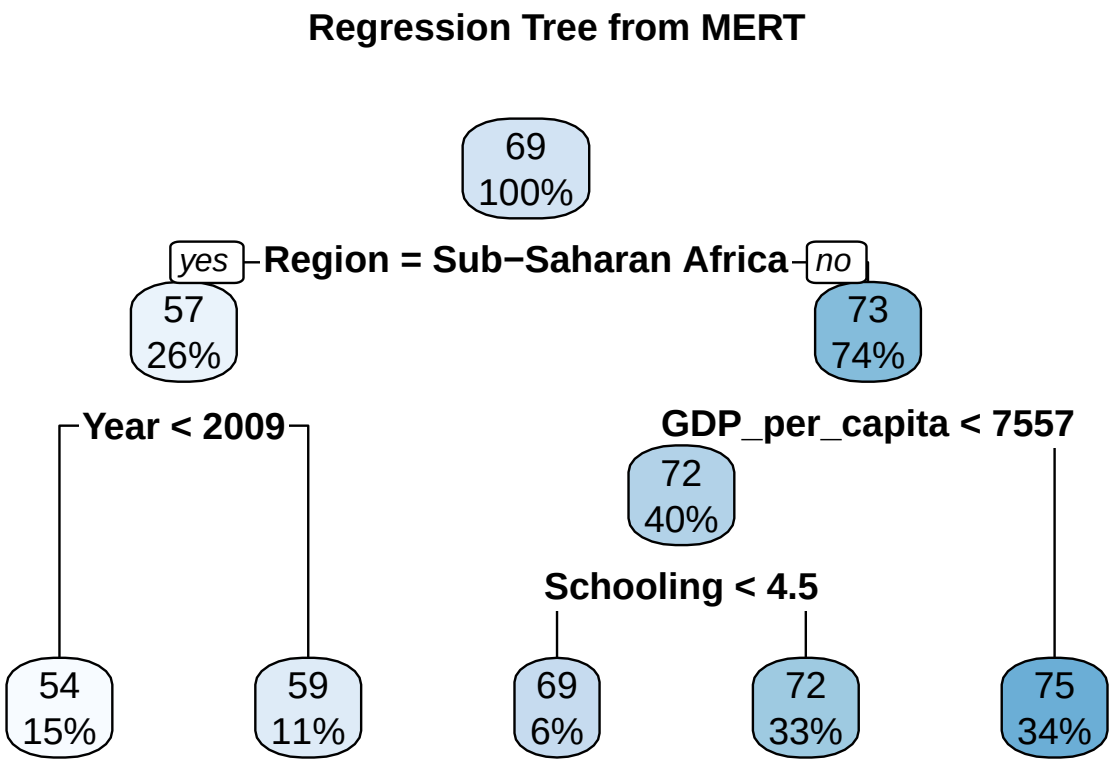
```
##   GLL at iteration 71 : 6031.676
## Iteration: 72
##   GLL at iteration 72 : 6031.676
## Iteration: 73
##   GLL at iteration 73 : 6031.675
## Iteration: 74
##   GLL at iteration 74 : 6031.675
## Iteration: 75
##   GLL at iteration 75 : 6031.675
## Iteration: 76
##   GLL at iteration 76 : 6031.674
## Iteration: 77
##   GLL at iteration 77 : 6031.674
## Iteration: 78
##   GLL at iteration 78 : 6031.674
## Iteration: 79
##   GLL at iteration 79 : 6031.674
## Iteration: 80
##   GLL at iteration 80 : 6031.674
## Iteration: 81
##   GLL at iteration 81 : 6031.674
## Iteration: 82
##   GLL at iteration 82 : 6031.674
## Iteration: 83
##   GLL at iteration 83 : 6031.673
## Iteration: 84
##   GLL at iteration 84 : 6031.673
##   Convergence reached at iteration 84
```

```
# look at final LME
summary(mert_result$lme_model)
```

```
## Linear mixed-effects model fit by REML
##   Data: data
##       AIC      BIC    logLik
##   11635.9 11653.78 -5814.952
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept) Residual
## StdDev:    4.744735 1.577554
##
## Fixed effects:  adjusted_y ~ 1
##                      Value Std.Error   DF       t-value p-value
## (Intercept) -4.252668e-14 0.3558612 2685 -1.195036e-13       1
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -5.11637670 -0.60350594  0.00976536  0.61850606  4.99194823
##
## Number of Observations: 2864
## Number of Groups: 179
```

```r
library(rpart.plot)
rpart.plot(mert_result$tree_model,
           type = 2,
           fallen.leaves = TRUE,
           cex = 1.2,
           main = "Regression Tree from MERT")
```

```
## Warning: Cannot retrieve the data used to build the model (so cannot determine roundint and is.binary
## To silence this warning:
##     Call rpart.plot with roundint=FALSE,
##     or rebuild the rpart model with model=TRUE.
```



3.2 MERF model

```r
MERF <- function(data, response, fixed_effects, cluster, max_iter = 150, tol = 1e-4, ntree = 100) {
  library(nlme)
  library(randomForest)

  data <- data.frame(data)
  data$random_effect <- 0
  GLL_history <- numeric(max_iter)

  for (iter in 1:max_iter) {
    cat("Iteration:", iter, "\n")

    # subtract random effects
```

```r
data$adjusted_response <- data[[response]] - data$random_effect

# random forest to adjusted response
rf_model <- randomForest(
  x = data[, fixed_effects],
  y = data$adjusted_response,
  ntree = ntree
)
data$fixed_pred <- predict(rf_model, newdata = data[, fixed_effects])

# randome part
data$adjusted_y <- data[[response]] - data$fixed_pred

# fit on randome part
lme_model <- tryCatch({
  lme(adjusted_y ~ 1,
      random = as.formula(paste("~ 1 |", cluster)),
      data = data,
      na.action = na.omit,
      control = lmeControl(opt = "optim"))
}, error = function(e) {
  cat("  LME model failed to converge at iteration", iter, "\n")
  return(NULL)
})
if (is.null(lme_model)) break

# update random effect
re <- ranef(lme_model)
re_vector <- re[as.character(data[[cluster]]), "(Intercept)"]
data$random_effect <- re_vector

# gll
sigma2 <- lme_model$sigma^2
tau2 <- as.numeric(VarCorr(lme_model)[1, "Variance"])
cluster_ids <- unique(data[[cluster]])
GLL <- 0
for (cid in cluster_ids) {
  subset <- data[data[[cluster]] == cid, ]
  y_i <- subset[[response]]
  f_Xi <- subset$fixed_pred
  b_i <- unique(subset$random_effect)  # scalar
  Z_i <- matrix(1, nrow = length(y_i), ncol = 1)
  R_i <- sigma2 * diag(length(y_i))
  D <- matrix(tau2, 1, 1)

  resid <- y_i - f_Xi - Z_i %*% b_i
  term1 <- t(resid) %*% solve(R_i) %*% resid
  term2 <- t(b_i) %*% solve(D) %*% b_i
  term3 <- log(det(D))
  term4 <- log(det(R_i))

  GLL <- GLL + term1 + term2 + term3 + term4
}
```

```
    GLL_history[iter] <- GLL
    cat("  GLL at iteration", iter, ":", GLL, "\n")

    # check convergence
    if (iter > 1 && abs(GLL_history[iter] - GLL_history[iter - 1]) < tol) {
      cat("  Convergence reached at iteration", iter, "\n")
      break
    }
  }
  return(list(
    rf_model = rf_model,
    lme_model = lme_model,
    final_data = data,
    GLL = GLL_history[1:iter]
  ))
}
```

```
fixed_vars <- c("Year", "Region", "Alcohol_consumption", "Hepatitis_B", "Measles", "BMI",
                "Polio", "Incidents_HIV", "GDP_per_capita", "Population_mln",
                "Thinness_ten_nineteen_years", "Schooling", "Economy_status_Developing")
merf_result <- MERF(
  data = life_final,
  response = "Life_expectancy",
  fixed_effects = fixed_vars,
  cluster = "Country",
  max_iter = 150,
  ntree = 100
)
```

```
## Iteration: 1
##   GLL at iteration 1 : -4811.845
## Iteration: 2
##   GLL at iteration 2 : -4671.191
## Iteration: 3
##   GLL at iteration 3 : -4506.071
## Iteration: 4
##   GLL at iteration 4 : -4680.02
## Iteration: 5
##   GLL at iteration 5 : -4734.982
## Iteration: 6
##   GLL at iteration 6 : -4833.171
## Iteration: 7
##   GLL at iteration 7 : -4669.601
## Iteration: 8
##   GLL at iteration 8 : -4727.141
## Iteration: 9
##   GLL at iteration 9 : -4859.512
## Iteration: 10
##   GLL at iteration 10 : -4879.402
## Iteration: 11
##   GLL at iteration 11 : -4922.793
## Iteration: 12
##   GLL at iteration 12 : -4809.653
```

```
## Iteration: 13
##   GLL at iteration 13 : -4950.602
## Iteration: 14
##   GLL at iteration 14 : -4869.116
## Iteration: 15
##   GLL at iteration 15 : -4841.462
## Iteration: 16
##   GLL at iteration 16 : -4990.477
## Iteration: 17
##   GLL at iteration 17 : -4890.857
## Iteration: 18
##   GLL at iteration 18 : -4879.695
## Iteration: 19
##   GLL at iteration 19 : -4975.291
## Iteration: 20
##   GLL at iteration 20 : -4943.455
## Iteration: 21
##   GLL at iteration 21 : -4989.286
## Iteration: 22
##   GLL at iteration 22 : -5098.913
## Iteration: 23
##   GLL at iteration 23 : -4925.46
## Iteration: 24
##   GLL at iteration 24 : -4917.093
## Iteration: 25
##   GLL at iteration 25 : -4942.132
## Iteration: 26
##   GLL at iteration 26 : -4995.503
## Iteration: 27
##   GLL at iteration 27 : -4970.922
## Iteration: 28
##   GLL at iteration 28 : -5011.225
## Iteration: 29
##   GLL at iteration 29 : -5003.379
## Iteration: 30
##   GLL at iteration 30 : -4954.282
## Iteration: 31
##   GLL at iteration 31 : -5101.491
## Iteration: 32
##   GLL at iteration 32 : -4998.113
## Iteration: 33
##   GLL at iteration 33 : -5079.777
## Iteration: 34
##   GLL at iteration 34 : -4885.084
## Iteration: 35
##   GLL at iteration 35 : -5028.573
## Iteration: 36
##   GLL at iteration 36 : -4953.445
## Iteration: 37
##   GLL at iteration 37 : -4938.381
## Iteration: 38
##   GLL at iteration 38 : -5037.969
## Iteration: 39
##   GLL at iteration 39 : -5007.729
```

```
## Iteration: 40
##   GLL at iteration 40 : -5038.025
## Iteration: 41
##   GLL at iteration 41 : -5030.64
## Iteration: 42
##   GLL at iteration 42 : -4918.511
## Iteration: 43
##   GLL at iteration 43 : -4960.189
## Iteration: 44
##   GLL at iteration 44 : -4987.087
## Iteration: 45
##   GLL at iteration 45 : -4876.547
## Iteration: 46
##   GLL at iteration 46 : -5161.557
## Iteration: 47
##   GLL at iteration 47 : -5039.369
## Iteration: 48
##   GLL at iteration 48 : -4953.75
## Iteration: 49
##   GLL at iteration 49 : -5033.264
## Iteration: 50
##   GLL at iteration 50 : -4890.507
## Iteration: 51
##   GLL at iteration 51 : -4990.878
## Iteration: 52
##   GLL at iteration 52 : -5143.449
## Iteration: 53
##   GLL at iteration 53 : -5077.265
## Iteration: 54
##   GLL at iteration 54 : -5138.485
## Iteration: 55
##   GLL at iteration 55 : -4999.679
## Iteration: 56
##   GLL at iteration 56 : -5063.716
## Iteration: 57
##   GLL at iteration 57 : -5074.143
## Iteration: 58
##   GLL at iteration 58 : -5093.061
## Iteration: 59
##   GLL at iteration 59 : -5010.786
## Iteration: 60
##   GLL at iteration 60 : -4916.915
## Iteration: 61
##   GLL at iteration 61 : -4984.523
## Iteration: 62
##   GLL at iteration 62 : -5026.362
## Iteration: 63
##   GLL at iteration 63 : -4999.599
## Iteration: 64
##   GLL at iteration 64 : -5163.027
## Iteration: 65
##   GLL at iteration 65 : -5098.68
## Iteration: 66
##   GLL at iteration 66 : -5145.91
```

```
## Iteration: 67
##   GLL at iteration 67 : -4998.447
## Iteration: 68
##   GLL at iteration 68 : -4966.897
## Iteration: 69
##   GLL at iteration 69 : -5103.282
## Iteration: 70
##   GLL at iteration 70 : -5061.846
## Iteration: 71
##   GLL at iteration 71 : -5156.347
## Iteration: 72
##   GLL at iteration 72 : -5063.733
## Iteration: 73
##   GLL at iteration 73 : -5123.295
## Iteration: 74
##   GLL at iteration 74 : -4971.992
## Iteration: 75
##   GLL at iteration 75 : -5069.776
## Iteration: 76
##   GLL at iteration 76 : -5016.308
## Iteration: 77
##   GLL at iteration 77 : -5118.705
## Iteration: 78
##   GLL at iteration 78 : -4911.496
## Iteration: 79
##   GLL at iteration 79 : -5081.21
## Iteration: 80
##   GLL at iteration 80 : -5143.896
## Iteration: 81
##   GLL at iteration 81 : -5007.716
## Iteration: 82
##   GLL at iteration 82 : -5135.296
## Iteration: 83
##   GLL at iteration 83 : -5042.636
## Iteration: 84
##   GLL at iteration 84 : -5058.327
## Iteration: 85
##   GLL at iteration 85 : -5167.511
## Iteration: 86
##   GLL at iteration 86 : -5058.61
## Iteration: 87
##   GLL at iteration 87 : -5072.103
## Iteration: 88
##   GLL at iteration 88 : -5327.242
## Iteration: 89
##   GLL at iteration 89 : -4950.261
## Iteration: 90
##   GLL at iteration 90 : -5118.459
## Iteration: 91
##   GLL at iteration 91 : -4978.966
## Iteration: 92
##   GLL at iteration 92 : -5064.05
## Iteration: 93
##   GLL at iteration 93 : -5134.532
```

```
## Iteration: 94
##   GLL at iteration 94 : -4958.104
## Iteration: 95
##   GLL at iteration 95 : -5177.411
## Iteration: 96
##   GLL at iteration 96 : -5043.808
## Iteration: 97
##   GLL at iteration 97 : -5090.545
## Iteration: 98
##   GLL at iteration 98 : -4950.839
## Iteration: 99
##   GLL at iteration 99 : -5225.487
## Iteration: 100
##   GLL at iteration 100 : -4984.732
## Iteration: 101
##   GLL at iteration 101 : -5089.61
## Iteration: 102
##   GLL at iteration 102 : -5119.574
## Iteration: 103
##   GLL at iteration 103 : -5114.671
## Iteration: 104
##   GLL at iteration 104 : -4957.31
## Iteration: 105
##   GLL at iteration 105 : -5040.363
## Iteration: 106
##   GLL at iteration 106 : -5056.443
## Iteration: 107
##   GLL at iteration 107 : -5059.026
## Iteration: 108
##   GLL at iteration 108 : -5112.801
## Iteration: 109
##   GLL at iteration 109 : -5105.879
## Iteration: 110
##   GLL at iteration 110 : -5161.193
## Iteration: 111
##   GLL at iteration 111 : -4920.671
## Iteration: 112
##   GLL at iteration 112 : -5091.733
## Iteration: 113
##   GLL at iteration 113 : -5188.483
## Iteration: 114
##   GLL at iteration 114 : -5030.794
## Iteration: 115
##   GLL at iteration 115 : -5199.404
## Iteration: 116
##   GLL at iteration 116 : -5134.312
## Iteration: 117
##   GLL at iteration 117 : -5097.351
## Iteration: 118
##   GLL at iteration 118 : -4998.745
## Iteration: 119
##   GLL at iteration 119 : -5148.384
## Iteration: 120
##   GLL at iteration 120 : -5279.105
```

```
## Iteration: 121
##   GLL at iteration 121 : -5174.994
## Iteration: 122
##   GLL at iteration 122 : -5088.741
## Iteration: 123
##   GLL at iteration 123 : -5161.811
## Iteration: 124
##   GLL at iteration 124 : -5198.182
## Iteration: 125
##   GLL at iteration 125 : -5212.371
## Iteration: 126
##   GLL at iteration 126 : -5057.161
## Iteration: 127
##   GLL at iteration 127 : -5145.998
## Iteration: 128
##   GLL at iteration 128 : -5235.651
## Iteration: 129
##   GLL at iteration 129 : -5195.462
## Iteration: 130
##   GLL at iteration 130 : -5057.16
## Iteration: 131
##   GLL at iteration 131 : -5030.603
## Iteration: 132
##   GLL at iteration 132 : -5070.934
## Iteration: 133
##   GLL at iteration 133 : -5100.717
## Iteration: 134
##   GLL at iteration 134 : -5023.492
## Iteration: 135
##   GLL at iteration 135 : -5091.637
## Iteration: 136
##   GLL at iteration 136 : -5043.071
## Iteration: 137
##   GLL at iteration 137 : -5132.256
## Iteration: 138
##   GLL at iteration 138 : -5151.566
## Iteration: 139
##   GLL at iteration 139 : -5056.806
## Iteration: 140
##   GLL at iteration 140 : -5074.404
## Iteration: 141
##   GLL at iteration 141 : -5115.01
## Iteration: 142
##   GLL at iteration 142 : -5226.971
## Iteration: 143
##   GLL at iteration 143 : -5289.793
## Iteration: 144
##   GLL at iteration 144 : -5044.183
## Iteration: 145
##   GLL at iteration 145 : -5063.248
## Iteration: 146
##   GLL at iteration 146 : -5171.536
## Iteration: 147
##   GLL at iteration 147 : -5344.241
```

```
## Iteration: 148
##   GLL at iteration 148 : -5204.674
## Iteration: 149
##   GLL at iteration 149 : -5054.869
## Iteration: 150
##   GLL at iteration 150 : -5087.955
```

```
summary(merf_result$lme_model)
```

```
## Linear mixed-effects model fit by REML
##   Data: data
##       AIC      BIC    logLik
##   1205.78 1223.659 -599.8902
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept)  Residual
## StdDev:    3.896678 0.2292037
##
## Fixed effects:  adjusted_y ~ 1
##                   Value Std.Error   DF     t-value p-value
## (Intercept) 0.002309529 0.2912829 2685 0.007928819  0.9937
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -9.679428879 -0.324829347  0.006758151  0.362981683  8.430950656
##
## Number of Observations: 2864
## Number of Groups: 179
```

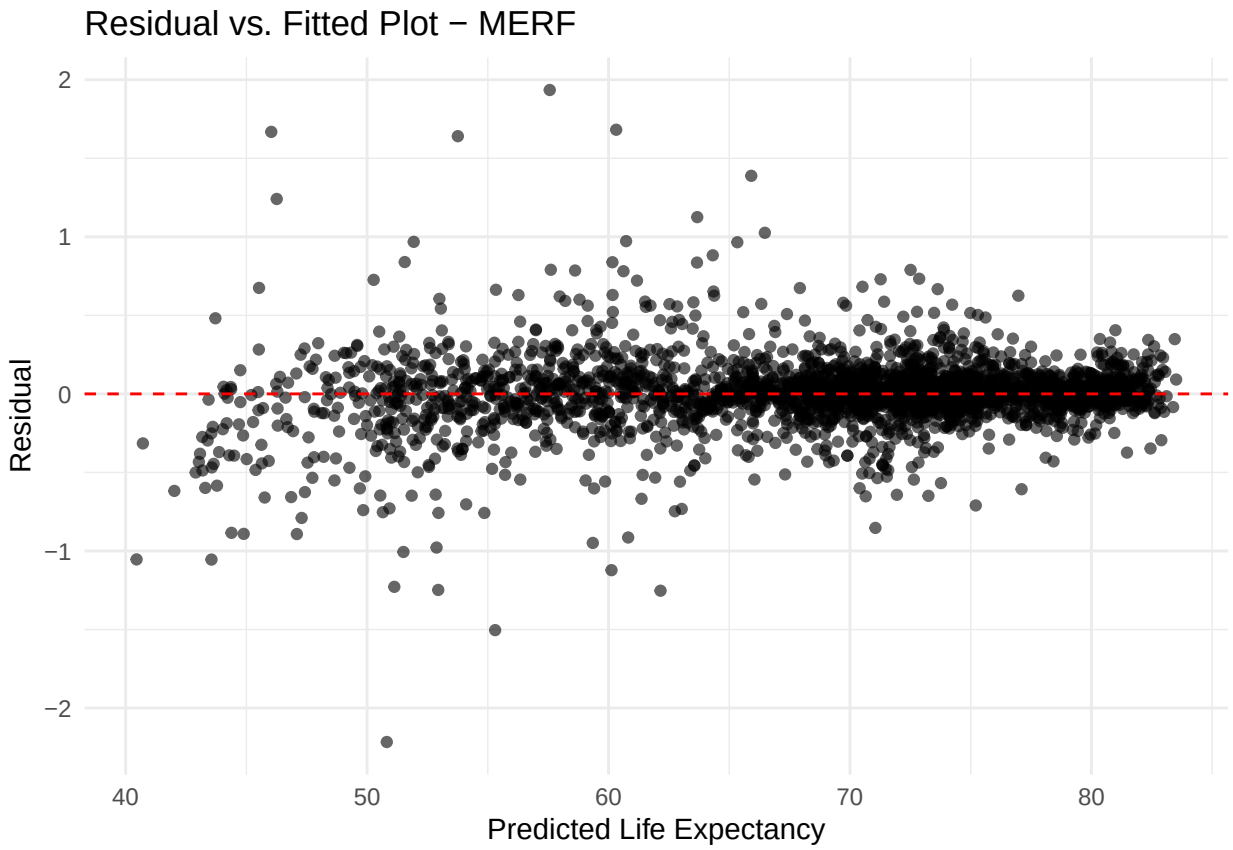
```
final_df_merf <- merf_result$final_data

final_df_merf$predicted_LE <- final_df_merf$fixed_pred + final_df_merf$random_effect

final_df_merf$residual <- final_df_merf$Life_expectancy - final_df_merf$predicted_LE

library(ggplot2)

ggplot(final_df_merf, aes(x = predicted_LE, y = residual)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(
    title = "Residual vs. Fitted Plot - MERF",
    x = "Predicted Life Expectancy",
    y = "Residual"
  ) +
  theme_minimal()
```



4. Error Evaluation with 10-fold CV

4.1 On the global dataset

```r
cv_linear_model <- function(formula, data, k = 10) {
  # create k folds
  set.seed(123)  # For reproducibility
  folds <- createFolds(data[[all.vars(formula)[1]]], k = k, list = TRUE)

  # vector to store MSE for each fold
  mse_values <- numeric(k)

  # k-fold cross-validation
  for (i in 1:k) {
    # training and validation sets
    training_set <- data[-folds[[i]], ]
    validation_set <- data[folds[[i]], ]

    # Fit the linear model on the training set
    lm_model <- lm(formula, data = training_set)

    # Predict on the validation set
    predictions <- predict(lm_model, newdata = validation_set)
```

```
    # Compute mse
    mse <- mean((validation_set[[all.vars(formula)[1]]] - predictions)^2)
    mse_values[i] <- mse
  }

  # average MSE across all folds
  cv_mse <- mean(mse_values)
  return(cv_mse)
}
```

```
linmod_cv <- cv_linear_model(Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measl
                                  Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_ninet
                                  Schooling + Economy_status_Developing,
              data = life_final, k = 10)
linmod_cv
```

```
## [1] 11.06292
```

```
library(nlme)
library(caret)

cv_lme <- function(formula, cluster, data, k = 10) {
  set.seed(123)
  folds <- createFolds(data[[all.vars(formula)[1]]], k = k, list = TRUE)
  mse_values <- numeric(k)

  for (i in 1:k) {
    train_set <- data[-folds[[i]], ]
    test_set <- data[folds[[i]], ]

    model <- lme(formula, random = as.formula(paste("~ 1 |", cluster)), data = train_set, na.action = na
    predictions <- predict(model, newdata = test_set)
    mse_values[i] <- mean((test_set[[all.vars(formula)[1]]] - predictions)^2)
  }
  return(mean(mse_values))
}
```

```
lme_cv <- cv_lme(
  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + BMI +
    Polio + Incidents_HIV + GDP_per_capita + Population_mln +
    Thinness_ten_nineteen_years + Schooling + Economy_status_Developing,
  cluster = "Country",
  data = life_final
)
lme_cv
```

```
## [1] 1.170252
```

```
cv_mert <- function(data, response, fixed_effects, cluster, k = 10) {
  set.seed(123)
  folds <- createFolds(data[[response]], k = k, list = TRUE)
  mse_values <- numeric(k)
```

```
  for (i in 1:k) {
    train <- data[-folds[[i]], ]
    test  <- data[folds[[i]], ]

    # Train MERT model on training data
    result <- MERT(train, response, fixed_effects, cluster)

    # Predict fixed effects from the trained tree on the test data
    test$fixed_pred <- predict(result$tree_model, newdata = test)

    # Predict random effects: match clusters in test to train
    test$random_effect <- 0  # default
    re <- ranef(result$lme_model)
    test_clusters <- as.character(test[[cluster]])
    re_vector <- re[test_clusters, "(Intercept)"]
    test$random_effect[!is.na(re_vector)] <- re_vector[!is.na(re_vector)]

    # Combine to get full prediction
    pred <- test$fixed_pred + test$random_effect

    # Compute MSE
    mse_values[i] <- mean((test[[response]] - pred)^2)
  }

  return(mean(mse_values))
}
```

```
invisible(capture.output(
  mert_cv_global <- cv_mert(
    data = life_final,
    response = "Life_expectancy",
    fixed_effects = fixed_vars,
    cluster = "Country"
  )
))
```

```
mert_cv_global
```

```
## [1] 2.718669
```

```
cv_merf <- function(data, response, fixed_effects, cluster, k = 10, ntree = 100) {
  set.seed(123)
  folds <- createFolds(data[[response]], k = k, list = TRUE)
  mse_values <- numeric(k)

  for (i in 1:k) {
    train <- data[-folds[[i]], ]
    test <- data[folds[[i]], ]

    result <- MERF(train, response, fixed_effects, cluster, ntree = ntree)

    # Predict fixed effects from RF on test data
```

```
    test$fixed_pred <- predict(result$rf_model, newdata = test[, fixed_effects])

    # Predict random effects using test's cluster info
    re <- ranef(result$lme_model)
    test$random_effect <- re[as.character(test[[cluster]]), "(Intercept)"]
    test$random_effect[is.na(test$random_effect)] <- 0  # handle unseen clusters

    # Final prediction
    pred <- test$fixed_pred + test$random_effect

    mse_values[i] <- mean((test[[response]] - pred)^2)
  }

  return(mean(mse_values))
}
```

```
invisible(capture.output(
  merf_cv_global <- cv_merf(
    data = life_final,
    response = "Life_expectancy",
    fixed_effects = fixed_vars,
    cluster = "Country",
    ntree = 100)
  ))
```

```
merf_cv_global
```

```
## [1] 0.2760915
```

4.2 On the SSA region dataset

```
## getting the SSA dataset
SSA = life_final[life_final$Region == "Sub-Saharan Africa",]
```

```
linmod_cv_SSA <- cv_linear_model(Life_expectancy ~ Year + Alcohol_consumption + Hepatitis_B + Measles +
                data = SSA, k = 10)
linmod_cv_SSA
```

```
## [1] 11.08265
```

```
lme_cv_SSA <- cv_lme(
  Life_expectancy ~ Year + Alcohol_consumption + Hepatitis_B + Measles + BMI +
    Polio + Incidents_HIV + GDP_per_capita + Population_mln +
    Thinness_ten_nineteen_years + Schooling,
  cluster = "Country",
  data = SSA
)
lme_cv_SSA
```

```
## [1] 2.329219
```

```
fixed_vars_no_region <- c("Year", "Alcohol_consumption", "Hepatitis_B", "Measles", "BMI",
                "Polio", "Incidents_HIV", "GDP_per_capita", "Population_mln",
                "Thinness_ten_nineteen_years", "Schooling")
```

```
invisible(capture.output(mert_cv_SSA <- cv_mert(
  data = SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars_no_region,
  cluster = "Country"
)))
```

```
mert_cv_SSA
```

```
## [1] 2.679867
```

```
invisible(capture.output(
  merf_cv_SSA <- cv_merf(
  data = SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars_no_region,
  cluster = "Country",
  ntree = 100
)))
```

```
merf_cv_SSA
```

```
## [1] 0.72034
```

4.3 On the non-SSA region dataset

```
## getting the SSA dataset
non_SSA = life_final[life_final$Region != "Sub-Saharan Africa",]
```

```
linmod_cv_non_SSA <- cv_linear_model(Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_
                data = non_SSA, k = 10)
linmod_cv_non_SSA
```

```
## [1] 7.724655
```

```
lme_cv_non_SSA <- cv_lme(
  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + BMI +
    Polio + Incidents_HIV + GDP_per_capita + Population_mln +
    Thinness_ten_nineteen_years + Schooling + Economy_status_Developing,
  cluster = "Country",
  data = non_SSA
)
lme_cv_non_SSA
```

```
## [1] 0.4811008
```

```r
invisible(capture.output(mert_cv_non_SSA <- cv_mert(
  data = non_SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars,
  cluster = "Country"
)))
```

```r
mert_cv_non_SSA
```

```
## [1] 0.8149134
```

```r
invisible(capture.output(merf_cv_non_SSA <- cv_merf(
  data = non_SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars,
  cluster = "Country",
  ntree = 100
)))
```

```r
merf_cv_non_SSA
```

```
## [1] 0.1285327
```

```r
# Combine all the results into a data frame
cv_results <- data.frame(
  Model = c("MLR", "LME(intercept)", "MERT", "MERF"),
  `Global Data` = c(
    round(linmod_cv, 2),
    round(lme_cv, 2),
    round(mert_cv_global, 2),
    round(merf_cv_global, 2)
  ),
  `SSA` = c(
    round(linmod_cv_SSA, 2),
    round(lme_cv_SSA, 2),
    round(mert_cv_SSA, 2),
    round(merf_cv_SSA, 2)
  ),
  `Non-SSA` = c(
    round(linmod_cv_non_SSA, 2),
    round(lme_cv_non_SSA, 2),
    round(mert_cv_non_SSA, 2),
    round(merf_cv_non_SSA, 2)
  )
)

knitr::kable(cv_results, caption = "10-Fold Cross-Validation Scores for Each Model")
```

Table 1: 10-Fold Cross-Validation Scores for Each Model

Model	Global.Data	SSA	Non.SSA
MLR	11.06	11.08	7.72
LME(intercept)	1.17	2.33	0.48
MERT	2.72	2.68	0.81
MERF	0.28	0.72	0.13

5. Interpretation

5.1 Global Data

```
lme_int <- lme(fixed = Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Measles + I
               random = ~ 1 | Country,
               data = life_final, na.action = na.omit)
summary(lme_int)
```

```
## Linear mixed-effects model fit by REML
##   Data: life_final
##        AIC     BIC    logLik
##   9415.956 9558.81 -4683.978
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept) Residual
## StdDev:    4.230153 1.033807
##
## Fixed effects:  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B +      Measles +
##                                         Value Std.Error   DF   t-value p-value
## (Intercept)                          -517.5989 21.039556 2674 -24.60123  0.0000
## Year                                    0.2950  0.011612 2674  25.40239  0.0000
## RegionCentral America and Caribbean     3.6697  1.334126  168   2.75061  0.0066
## RegionEuropean Union                   -1.3077  1.965739  168  -0.66524  0.5068
## RegionMiddle East                       4.4215  1.476299  168   2.99498  0.0032
## RegionNorth America                     2.0926  2.805876  168   0.74579  0.4568
## RegionOceania                          -0.8141  1.609603  168  -0.50577  0.6137
## RegionRest of Africa                    3.8956  2.299460  168   1.69414  0.0921
## RegionRest of Europe                    3.0517  1.467239  168   2.07992  0.0391
## RegionSouth America                     3.8007  1.515677  168   2.50763  0.0131
## RegionSub-Saharan Africa               -8.8460  1.069721  168  -8.26945  0.0000
## Alcohol_consumption                     0.0526  0.024878 2674   2.11374  0.0346
## Hepatitis_B                             0.0025  0.002779 2674   0.91234  0.3617
## Measles                                -0.0129  0.002877 2674  -4.48901  0.0000
## BMI                                     0.0335  0.112677 2674   0.29721  0.7663
## Polio                                   0.0595  0.003358 2674  17.72030  0.0000
## Incidents_HIV                          -1.1226  0.034498 2674 -32.54128  0.0000
## GDP_per_capita                          0.0000  0.000012 2674   0.09386  0.9252
## Population_mln                          0.0029  0.001869 2674   1.54149  0.1233
## Thinness_ten_nineteen_years            -0.0192  0.011573 2674  -1.65829  0.0974
## Schooling                              -0.1524  0.056980 2674  -2.67426  0.0075
```

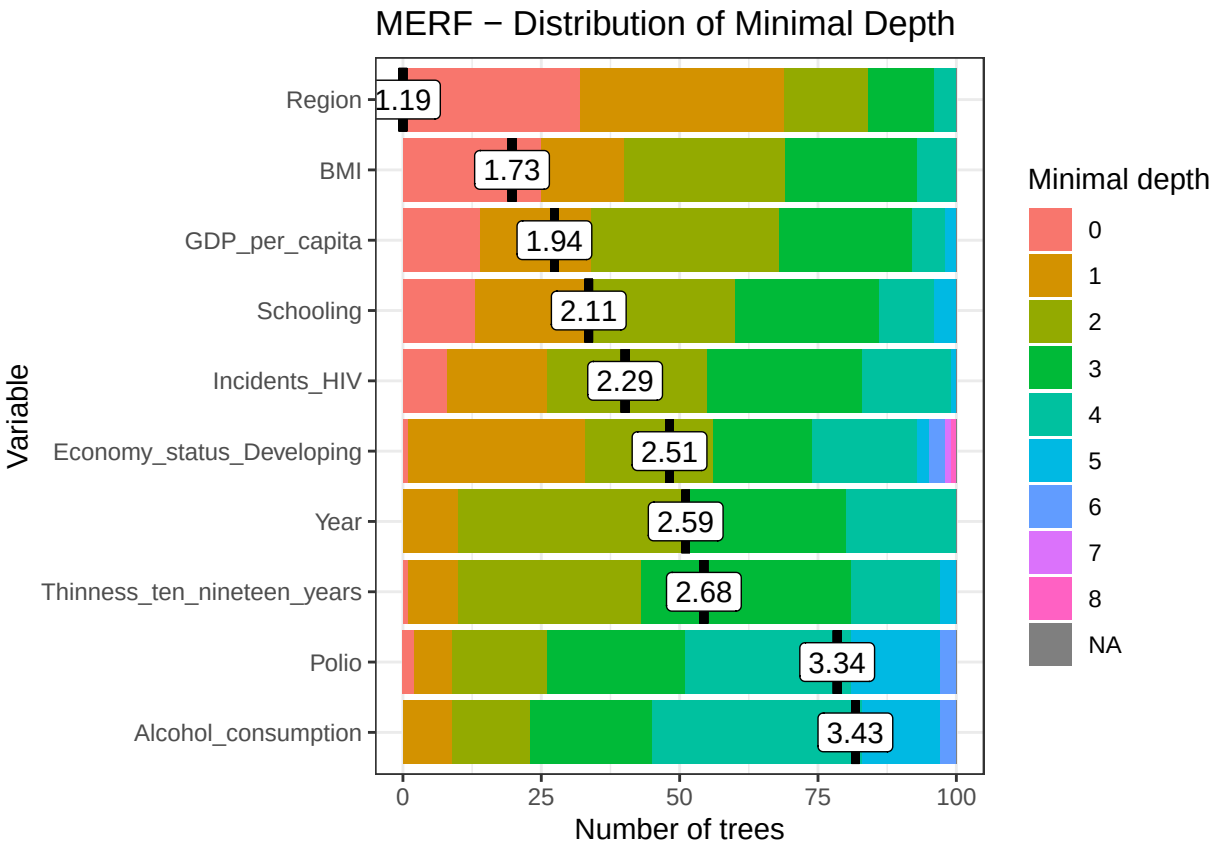
```
## Economy_status_Developing             -9.8183  1.665372  168  -5.89558  0.0000
##  Correlation:
##                                    (Intr) Year   RgCAaC RgnErU RgnMdE RgnNrA
## Year                               -0.994
## RegionCentral America and Caribbean -0.177  0.166
## RegionEuropean Union               -0.147  0.079  0.276
## RegionMiddle East                  -0.221  0.209  0.426  0.301
## RegionNorth America                -0.165  0.134  0.200  0.420  0.217
## RegionOceania                      -0.194  0.174  0.392  0.378  0.377  0.240
## RegionRest of Africa               -0.073  0.067  0.261  0.153  0.242  0.111
## RegionRest of Europe               -0.208  0.173  0.407  0.472  0.388  0.284
## RegionSouth America                -0.150  0.139  0.403  0.238  0.365  0.174
## RegionSub-Saharan Africa            0.130 -0.154  0.495  0.297  0.430  0.189
## Alcohol_consumption                -0.136  0.136 -0.042 -0.060  0.063  0.002
## Hepatitis_B                         0.081 -0.079 -0.011 -0.008 -0.005  0.011
## Measles                             0.039 -0.041 -0.019 -0.015 -0.024 -0.012
## BMI                                 0.700 -0.745 -0.223 -0.131 -0.273 -0.161
## Polio                               0.050 -0.047  0.001  0.009  0.002  0.005
## Incidents_HIV                      -0.218  0.210  0.005 -0.005  0.030  0.013
## GDP_per_capita                      0.182 -0.197 -0.039  0.104 -0.114 -0.018
## Population_mln                      0.111 -0.118  0.176  0.118  0.129 -0.014
## Thinness_ten_nineteen_years         0.016 -0.021  0.055  0.032  0.020  0.021
## Schooling                           0.480 -0.451 -0.001 -0.007  0.019 -0.018
## Economy_status_Developing           0.028 -0.097 -0.054  0.769  0.011  0.334
##                                    RgnOcn RgnRoA RgnRoE RgnStA RgS-SA Alchl_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa                0.221
## RegionRest of Europe                0.389  0.223
## RegionSouth America                 0.337  0.224  0.352
## RegionSub-Saharan Africa            0.399  0.289  0.420  0.429
## Alcohol_consumption                 0.019  0.028 -0.053 -0.041 -0.053
## Hepatitis_B                        -0.005 -0.005  0.002 -0.010 -0.004  0.043
## Measles                            -0.002 -0.026 -0.026 -0.001  0.010 -0.051
## BMI                                -0.256 -0.115 -0.179 -0.181  0.058 -0.126
## Polio                               0.020 -0.003 -0.006  0.004  0.021 -0.071
## Incidents_HIV                       0.013  0.008  0.011  0.004 -0.106  0.034
## GDP_per_capita                     -0.006 -0.017 -0.021 -0.024  0.041 -0.114
## Population_mln                      0.141  0.088  0.143  0.122  0.212 -0.057
## Thinness_ten_nineteen_years         0.046  0.015  0.050  0.055  0.047 -0.077
## Schooling                           0.018  0.039 -0.080 -0.011  0.148 -0.021
## Economy_status_Developing           0.140 -0.022  0.219 -0.046 -0.025  0.023
##                                    Hptt_B Measls BMI    Polio  In_HIV GDP_p_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
```

```
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles                              -0.193
## BMI                                   0.027  0.006
## Polio                                -0.354 -0.058 -0.077
## Incidents_HIV                        -0.001 -0.060 -0.099  0.100
## GDP_per_capita                       -0.023 -0.008  0.114  0.051 -0.101
## Population_mln                        0.005 -0.025  0.058 -0.073 -0.035  0.041
## Thinness_ten_nineteen_years           0.019 -0.001  0.025 -0.040 -0.142 -0.008
## Schooling                            -0.041 -0.035 -0.055  0.075 -0.014 -0.113
## Economy_status_Developing            -0.012 -0.001  0.007  0.029 -0.026  0.271
##                                      Ppltn_ Thn___ Schlng
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## RegionSub-Saharan Africa
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI
## Polio
## Incidents_HIV
## GDP_per_capita
## Population_mln
## Thinness_ten_nineteen_years           0.041
## Schooling                             0.010  0.056
## Economy_status_Developing             0.012 -0.020  0.088
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -6.11484506 -0.42295445 -0.02410951  0.40086603  8.93794278
##
## Number of Observations: 2864
## Number of Groups: 179
```

```r
# Extract RF model from MERF model fit previously
rf_merf <- merf_result$rf_model

# Compute minimal depth distribution
min_depth_merf <- min_depth_distribution(rf_merf)

# Plot
plot_min_depth_distribution(min_depth_merf, mean_sample = "all_trees") +
  ggtitle("MERF - Distribution of Minimal Depth")
```



5.2 SSA region

```
# fit LME to SSA
lme_int_SSA <- lme(fixed = Life_expectancy ~ Year + Alcohol_consumption + Hepatitis_B + Measles + BMI +
                   random = ~ 1 | Country,
                   data = SSA, na.action = na.omit)
summary(lme_int_SSA)
```

```
## Linear mixed-effects model fit by REML
##   Data: SSA
##        AIC      BIC    logLik
##   3066.339 3130.832 -1519.169
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept) Residual
## StdDev:    9.851781 1.432088
##
## Fixed effects:  Life_expectancy ~ Year + Alcohol_consumption + Hepatitis_B +      Measles + BMI + Po
##                                  Value Std.Error  DF    t-value p-value
## (Intercept)                -1188.6709  74.84068 694 -15.882684  0.0000
## Year                           0.6475   0.04160 694  15.564927  0.0000
## Alcohol_consumption            0.1955   0.06732 694   2.903434  0.0038
## Hepatitis_B                    0.0044   0.00982 694   0.446398  0.6554
## Measles                       -0.0835   0.01525 694  -5.476924  0.0000
```

```
## BMI                            -2.1545   0.46494 694  -4.634029  0.0000
## Polio                           0.0347   0.00744 694   4.661441  0.0000
## Incidents_HIV                  -0.7931   0.06000 694 -13.217937  0.0000
## GDP_per_capita                 -0.0002   0.00008 694  -2.829629  0.0048
## Population_mln                  0.0759   0.01644 694   4.614311  0.0000
## Thinness_ten_nineteen_years     0.0353   0.02279 694   1.549053  0.1218
## Schooling                      -0.4395   0.20971 694  -2.095727  0.0365
##  Correlation:
##                               (Intr) Year   Alchl_ Hptt_B Measls BMI    Polio
## Year                          -0.997
## Alcohol_consumption           -0.097  0.096
## Hepatitis_B                    0.037 -0.046  0.136
## Measles                        0.096 -0.107  0.020 -0.049
## BMI                            0.743 -0.792 -0.096  0.067  0.064
## Polio                          0.221 -0.196 -0.185 -0.332  0.024 -0.047
## Incidents_HIV                 -0.368  0.353  0.064  0.075  0.049 -0.206 -0.115
## GDP_per_capita                 0.176 -0.165 -0.094  0.018 -0.005  0.051  0.067
## Population_mln                 0.227 -0.209  0.179 -0.089  0.030  0.016  0.004
## Thinness_ten_nineteen_years    0.002  0.008 -0.133 -0.029 -0.048 -0.102  0.056
## Schooling                      0.234 -0.193 -0.034 -0.156  0.069 -0.227  0.122
##                               In_HIV GDP_p_ Ppltn_ Thn___
## Year
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI
## Polio
## Incidents_HIV
## GDP_per_capita                -0.089
## Population_mln                -0.191  0.050
## Thinness_ten_nineteen_years   -0.144 -0.082  0.235
## Schooling                      0.183 -0.086 -0.032  0.131
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -4.13208079 -0.53509863 -0.05114064  0.47045724  4.98666609
##
## Number of Observations: 752
## Number of Groups: 47
```

```r
# fit MERF to SSA
merf_result_SSA <- MERF(
  data = SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars_no_region,
  cluster = "Country",
  max_iter = 150,
  ntree = 100
)
```

```
## Iteration: 1
##   GLL at iteration 1 : -474.1217
## Iteration: 2
##   GLL at iteration 2 : -440.876
```

```
## Iteration: 3
##   GLL at iteration 3 : -393.5427
## Iteration: 4
##   GLL at iteration 4 : -456.2412
## Iteration: 5
##   GLL at iteration 5 : -471.8431
## Iteration: 6
##   GLL at iteration 6 : -474.4348
## Iteration: 7
##   GLL at iteration 7 : -456.7133
## Iteration: 8
##   GLL at iteration 8 : -476.8689
## Iteration: 9
##   GLL at iteration 9 : -565.2019
## Iteration: 10
##   GLL at iteration 10 : -407.5486
## Iteration: 11
##   GLL at iteration 11 : -521.3936
## Iteration: 12
##   GLL at iteration 12 : -468.2732
## Iteration: 13
##   GLL at iteration 13 : -448.3483
## Iteration: 14
##   GLL at iteration 14 : -524.4245
## Iteration: 15
##   GLL at iteration 15 : -560.7817
## Iteration: 16
##   GLL at iteration 16 : -503.0129
## Iteration: 17
##   GLL at iteration 17 : -544.2224
## Iteration: 18
##   GLL at iteration 18 : -510.5982
## Iteration: 19
##   GLL at iteration 19 : -565.434
## Iteration: 20
##   GLL at iteration 20 : -508.4594
## Iteration: 21
##   GLL at iteration 21 : -472.1277
## Iteration: 22
##   GLL at iteration 22 : -576.2048
## Iteration: 23
##   GLL at iteration 23 : -574.6195
## Iteration: 24
##   GLL at iteration 24 : -584.7459
## Iteration: 25
##   GLL at iteration 25 : -512.8823
## Iteration: 26
##   GLL at iteration 26 : -538.156
## Iteration: 27
##   GLL at iteration 27 : -542.7959
## Iteration: 28
##   GLL at iteration 28 : -548.4734
## Iteration: 29
##   GLL at iteration 29 : -531.3512
```

```
## Iteration: 30
##   GLL at iteration 30 : -542.4552
## Iteration: 31
##   GLL at iteration 31 : -504.5714
## Iteration: 32
##   GLL at iteration 32 : -539.7953
## Iteration: 33
##   GLL at iteration 33 : -611.0166
## Iteration: 34
##   GLL at iteration 34 : -549.6814
## Iteration: 35
##   GLL at iteration 35 : -541.5566
## Iteration: 36
##   GLL at iteration 36 : -551.4486
## Iteration: 37
##   GLL at iteration 37 : -571.0567
## Iteration: 38
##   GLL at iteration 38 : -580.5374
## Iteration: 39
##   GLL at iteration 39 : -590.4959
## Iteration: 40
##   GLL at iteration 40 : -539.6001
## Iteration: 41
##   GLL at iteration 41 : -548.5665
## Iteration: 42
##   GLL at iteration 42 : -515.4033
## Iteration: 43
##   GLL at iteration 43 : -534.233
## Iteration: 44
##   GLL at iteration 44 : -514.7754
## Iteration: 45
##   GLL at iteration 45 : -534.2447
## Iteration: 46
##   GLL at iteration 46 : -563.0046
## Iteration: 47
##   GLL at iteration 47 : -517.2438
## Iteration: 48
##   GLL at iteration 48 : -574.6428
## Iteration: 49
##   GLL at iteration 49 : -545.7251
## Iteration: 50
##   GLL at iteration 50 : -610.8252
## Iteration: 51
##   GLL at iteration 51 : -513.6622
## Iteration: 52
##   GLL at iteration 52 : -582.1618
## Iteration: 53
##   GLL at iteration 53 : -596.0186
## Iteration: 54
##   GLL at iteration 54 : -580.2772
## Iteration: 55
##   GLL at iteration 55 : -600.2883
## Iteration: 56
##   GLL at iteration 56 : -528.5124
```

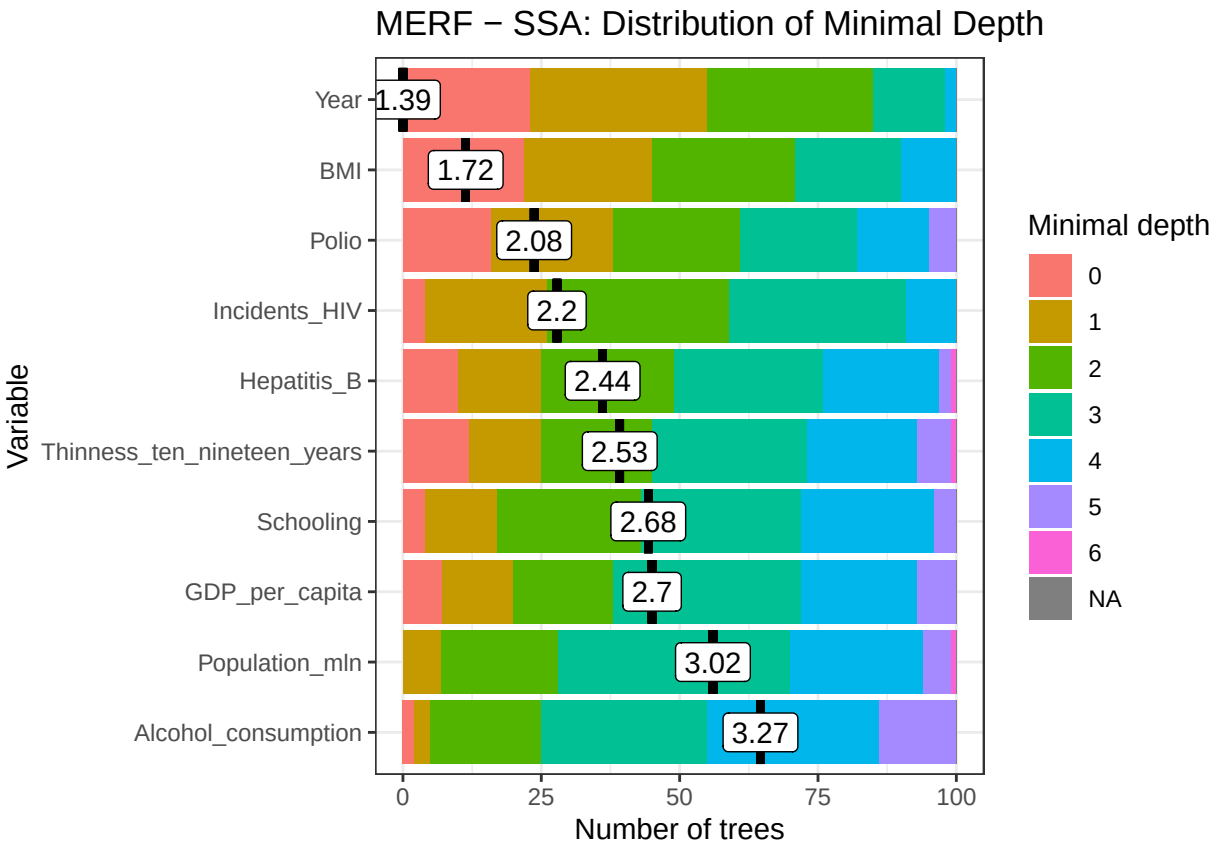
```
## Iteration: 57
##   GLL at iteration 57 : -577.0209
## Iteration: 58
##   GLL at iteration 58 : -574.9749
## Iteration: 59
##   GLL at iteration 59 : -594.4838
## Iteration: 60
##   GLL at iteration 60 : -576.0014
## Iteration: 61
##   GLL at iteration 61 : -536.3371
## Iteration: 62
##   GLL at iteration 62 : -620.7655
## Iteration: 63
##   GLL at iteration 63 : -573.0125
## Iteration: 64
##   GLL at iteration 64 : -560.8462
## Iteration: 65
##   GLL at iteration 65 : -564.3458
## Iteration: 66
##   GLL at iteration 66 : -615.4372
## Iteration: 67
##   GLL at iteration 67 : -580.0547
## Iteration: 68
##   GLL at iteration 68 : -607.399
## Iteration: 69
##   GLL at iteration 69 : -616.4299
## Iteration: 70
##   GLL at iteration 70 : -569.1469
## Iteration: 71
##   GLL at iteration 71 : -599.8213
## Iteration: 72
##   GLL at iteration 72 : -622.8112
## Iteration: 73
##   GLL at iteration 73 : -585.6656
## Iteration: 74
##   GLL at iteration 74 : -637.2368
## Iteration: 75
##   GLL at iteration 75 : -634.063
## Iteration: 76
##   GLL at iteration 76 : -623.1408
## Iteration: 77
##   GLL at iteration 77 : -521.7894
## Iteration: 78
##   GLL at iteration 78 : -623.7477
## Iteration: 79
##   GLL at iteration 79 : -619.9063
## Iteration: 80
##   GLL at iteration 80 : -649.1637
## Iteration: 81
##   GLL at iteration 81 : -560.824
## Iteration: 82
##   GLL at iteration 82 : -600.1887
## Iteration: 83
##   GLL at iteration 83 : -548.0202
```

```
## Iteration: 84
##   GLL at iteration 84 : -630.2165
## Iteration: 85
##   GLL at iteration 85 : -648.3967
## Iteration: 86
##   GLL at iteration 86 : -585.557
## Iteration: 87
##   GLL at iteration 87 : -603.2486
## Iteration: 88
##   GLL at iteration 88 : -575.0975
## Iteration: 89
##   GLL at iteration 89 : -660.6272
## Iteration: 90
##   GLL at iteration 90 : -582.2436
## Iteration: 91
##   GLL at iteration 91 : -588.0112
## Iteration: 92
##   GLL at iteration 92 : -562.1106
## Iteration: 93
##   GLL at iteration 93 : -615.5222
## Iteration: 94
##   GLL at iteration 94 : -615.4091
## Iteration: 95
##   GLL at iteration 95 : -576.9161
## Iteration: 96
##   GLL at iteration 96 : -621.2824
## Iteration: 97
##   GLL at iteration 97 : -570.0183
## Iteration: 98
##   GLL at iteration 98 : -576.6118
## Iteration: 99
##   GLL at iteration 99 : -617.2714
## Iteration: 100
##   GLL at iteration 100 : -615.8346
## Iteration: 101
##   GLL at iteration 101 : -573.5496
## Iteration: 102
##   GLL at iteration 102 : -616.0503
## Iteration: 103
##   GLL at iteration 103 : -581.6958
## Iteration: 104
##   GLL at iteration 104 : -629.2862
## Iteration: 105
##   GLL at iteration 105 : -613.0222
## Iteration: 106
##   GLL at iteration 106 : -578.7096
## Iteration: 107
##   GLL at iteration 107 : -594.3163
## Iteration: 108
##   GLL at iteration 108 : -584.0647
## Iteration: 109
##   GLL at iteration 109 : -555.7979
## Iteration: 110
##   GLL at iteration 110 : -621.2648
```

```
## Iteration: 111
##   GLL at iteration 111 : -604.9378
## Iteration: 112
##   GLL at iteration 112 : -614.6573
## Iteration: 113
##   GLL at iteration 113 : -612.4635
## Iteration: 114
##   GLL at iteration 114 : -597.4915
## Iteration: 115
##   GLL at iteration 115 : -597.0516
## Iteration: 116
##   GLL at iteration 116 : -660.1253
## Iteration: 117
##   GLL at iteration 117 : -617.7521
## Iteration: 118
##   GLL at iteration 118 : -641.0483
## Iteration: 119
##   GLL at iteration 119 : -565.5577
## Iteration: 120
##   GLL at iteration 120 : -632.0956
## Iteration: 121
##   GLL at iteration 121 : -580.4796
## Iteration: 122
##   GLL at iteration 122 : -654.5215
## Iteration: 123
##   GLL at iteration 123 : -647.13
## Iteration: 124
##   GLL at iteration 124 : -605.5589
## Iteration: 125
##   GLL at iteration 125 : -622.6028
## Iteration: 126
##   GLL at iteration 126 : -605.21
## Iteration: 127
##   GLL at iteration 127 : -602.5904
## Iteration: 128
##   GLL at iteration 128 : -608.9821
## Iteration: 129
##   GLL at iteration 129 : -607.9627
## Iteration: 130
##   GLL at iteration 130 : -606.9604
## Iteration: 131
##   GLL at iteration 131 : -619.6265
## Iteration: 132
##   GLL at iteration 132 : -616.9923
## Iteration: 133
##   GLL at iteration 133 : -621.9662
## Iteration: 134
##   GLL at iteration 134 : -654.5462
## Iteration: 135
##   GLL at iteration 135 : -616.9963
## Iteration: 136
##   GLL at iteration 136 : -658.5581
## Iteration: 137
##   GLL at iteration 137 : -600.9338
```

```
## Iteration: 138
##   GLL at iteration 138 : -594.2653
## Iteration: 139
##   GLL at iteration 139 : -626.8331
## Iteration: 140
##   GLL at iteration 140 : -683.3672
## Iteration: 141
##   GLL at iteration 141 : -650.2501
## Iteration: 142
##   GLL at iteration 142 : -640.3385
## Iteration: 143
##   GLL at iteration 143 : -617.1645
## Iteration: 144
##   GLL at iteration 144 : -631.3516
## Iteration: 145
##   GLL at iteration 145 : -611.9932
## Iteration: 146
##   GLL at iteration 146 : -643.6922
## Iteration: 147
##   GLL at iteration 147 : -612.3122
## Iteration: 148
##   GLL at iteration 148 : -627.2099
## Iteration: 149
##   GLL at iteration 149 : -609.8582
## Iteration: 150
##   GLL at iteration 150 : -625.1525
```

```r
# SSA
min_depth_ssa <- min_depth_distribution(merf_result_SSA$rf_model)
plot_min_depth_distribution(min_depth_ssa, mean_sample = "all_trees") +
  ggtitle("MERF - SSA: Distribution of Minimal Depth")
```



5.3 non-SSA region

```
lme_int_non_SSA <- lme(fixed = Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B + Mea
               random = ~ 1 | Country,
               data = non_SSA, na.action = na.omit)
summary(lme_int_non_SSA)
```

```
## Linear mixed-effects model fit by REML
##   Data: non_SSA
##        AIC      BIC    logLik
##   5162.781 5292.626 -2558.391
##
## Random effects:
##  Formula: ~1 | Country
##         (Intercept)  Residual
## StdDev:    3.274409 0.6653221
##
## Fixed effects:  Life_expectancy ~ Year + Region + Alcohol_consumption + Hepatitis_B +      Measles +
##                                        Value Std.Error   DF    t-value p-value
## (Intercept)                         -367.2345 16.152302 1969 -22.735736  0.0000
## Year                                   0.2165  0.008887 1969  24.364340  0.0000
## RegionCentral America and Caribbean    2.8271  1.031070  122   2.741920  0.0070
## RegionEuropean Union                  -1.5359  1.516072  122  -1.013094  0.3130
## RegionMiddle East                      3.0606  1.135941  122   2.694306  0.0080
```

```
## RegionNorth America                    0.9667  2.168698  122   0.445762  0.6566
## RegionOceania                         -1.7923  1.240067  122  -1.445298  0.1509
## RegionRest of Africa                   3.3137  1.776054  122   1.865785  0.0645
## RegionRest of Europe                   1.8552  1.132795  122   1.637740  0.1041
## RegionSouth America                    3.0683  1.171410  122   2.619282  0.0099
## Alcohol_consumption                    0.0338  0.019171 1969   1.760626  0.0785
## Hepatitis_B                            0.0058  0.001999 1969   2.896913  0.0038
## Measles                               -0.0015  0.001969 1969  -0.744370  0.4567
## BMI                                    0.2542  0.083617 1969   3.040214  0.0024
## Polio                                  0.0384  0.003125 1969  12.301585  0.0000
## Incidents_HIV                         -1.4283  0.201001 1969  -7.105905  0.0000
## GDP_per_capita                         0.0000  0.000008 1969   3.568659  0.0004
## Population_mln                         0.0019  0.001358 1969   1.363660  0.1728
## Thinness_ten_nineteen_years           -0.0154  0.011363 1969  -1.354108  0.1759
## Schooling                              0.0069  0.041445 1969   0.167148  0.8673
## Economy_status_Developing             -8.2435  1.277222  122  -6.454210  0.0000
##  Correlation:
##                                    (Intr) Year   RgCAaC RgnErU RgnMdE RgnNrA
## Year                               -0.994
## RegionCentral America and Caribbean -0.167  0.156
## RegionEuropean Union               -0.143  0.074  0.274
## RegionMiddle East                  -0.217  0.203  0.421  0.300
## RegionNorth America                -0.160  0.129  0.199  0.420  0.216
## RegionOceania                      -0.189  0.168  0.388  0.375  0.372  0.238
## RegionRest of Africa               -0.070  0.063  0.257  0.151  0.238  0.110
## RegionRest of Europe               -0.205  0.170  0.404  0.472  0.385  0.283
## RegionSouth America                -0.143  0.132  0.402  0.236  0.361  0.173
## Alcohol_consumption                -0.140  0.139 -0.044 -0.061  0.059  0.003
## Hepatitis_B                         0.127 -0.122 -0.004 -0.002 -0.007  0.008
## Measles                             0.065 -0.064 -0.011 -0.008 -0.023 -0.009
## BMI                                 0.701 -0.746 -0.219 -0.123 -0.262 -0.155
## Polio                              -0.082  0.077 -0.009 -0.003  0.009  0.008
## Incidents_HIV                      -0.096  0.076 -0.080 -0.025 -0.017 -0.021
## GDP_per_capita                      0.216 -0.228 -0.034  0.090 -0.108 -0.020
## Population_mln                      0.113 -0.119  0.166  0.114  0.119 -0.013
## Thinness_ten_nineteen_years         0.033 -0.040  0.053  0.029  0.013  0.021
## Schooling                           0.538 -0.512 -0.012 -0.013 -0.002 -0.027
## Economy_status_Developing           0.032 -0.102 -0.053  0.770  0.014  0.336
##                                    RgnOcn RgnRoA RgnRoE RgnStA Alchl_ Hptt_B
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa                0.217
## RegionRest of Europe                0.386  0.221
## RegionSouth America                 0.334  0.221  0.349
## Alcohol_consumption                 0.019  0.026 -0.055 -0.041
## Hepatitis_B                        -0.005 -0.003  0.002 -0.005  0.003
## Measles                            -0.002 -0.022 -0.020  0.004 -0.059 -0.167
## BMI                                -0.245 -0.106 -0.174 -0.176 -0.134  0.043
## Polio                               0.016 -0.006 -0.008 -0.005 -0.017 -0.402
## Incidents_HIV                      -0.039 -0.005 -0.011 -0.049 -0.003 -0.040
```

```
## GDP_per_capita                      -0.011 -0.017 -0.023 -0.022 -0.124 -0.012
## Population_mln                       0.130  0.082  0.136  0.115 -0.099  0.033
## Thinness_ten_nineteen_years          0.045  0.014  0.050  0.057  0.000  0.039
## Schooling                            0.001  0.029 -0.089 -0.021 -0.015  0.002
## Economy_status_Developing            0.139 -0.023  0.221 -0.046  0.023 -0.007
##                                    Measls BMI    Polio  In_HIV GDP_p_ Ppltn_
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI                                  0.009
## Polio                               -0.152 -0.109
## Incidents_HIV                       -0.068  0.125  0.071
## GDP_per_capita                       0.014  0.131  0.001 -0.074
## Population_mln                      -0.022  0.061 -0.117  0.006  0.044
## Thinness_ten_nineteen_years          0.038  0.063 -0.140 -0.053  0.014  0.022
## Schooling                           -0.027  0.010  0.011 -0.095 -0.084  0.015
## Economy_status_Developing            0.003  0.012  0.013 -0.040  0.241  0.009
##                                    Thn___ Schlng
## Year
## RegionCentral America and Caribbean
## RegionEuropean Union
## RegionMiddle East
## RegionNorth America
## RegionOceania
## RegionRest of Africa
## RegionRest of Europe
## RegionSouth America
## Alcohol_consumption
## Hepatitis_B
## Measles
## BMI
## Polio
## Incidents_HIV
## GDP_per_capita
## Population_mln
## Thinness_ten_nineteen_years
## Schooling                            0.034
## Economy_status_Developing           -0.019  0.093
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -5.771911412 -0.441738810  0.002433622  0.482275614  3.847935575
##
## Number of Observations: 2112
## Number of Groups: 132
```

```r
merf_result_non_SSA <- MERF(
  data = non_SSA,
  response = "Life_expectancy",
  fixed_effects = fixed_vars,
  cluster = "Country",
  max_iter = 150,
  ntree = 100
)
```

```
## Iteration: 1
##   GLL at iteration 1 : -4596.553
## Iteration: 2
##   GLL at iteration 2 : -4530.744
## Iteration: 3
##   GLL at iteration 3 : -4559.812
## Iteration: 4
##   GLL at iteration 4 : -4629.96
## Iteration: 5
##   GLL at iteration 5 : -4687.142
## Iteration: 6
##   GLL at iteration 6 : -4687.663
## Iteration: 7
##   GLL at iteration 7 : -4792.877
## Iteration: 8
##   GLL at iteration 8 : -4644.84
## Iteration: 9
##   GLL at iteration 9 : -4585.613
## Iteration: 10
##   GLL at iteration 10 : -4810.721
## Iteration: 11
##   GLL at iteration 11 : -4825.431
## Iteration: 12
##   GLL at iteration 12 : -4759.886
## Iteration: 13
##   GLL at iteration 13 : -4793.565
## Iteration: 14
##   GLL at iteration 14 : -4863.656
## Iteration: 15
##   GLL at iteration 15 : -4876.917
## Iteration: 16
##   GLL at iteration 16 : -4872.856
## Iteration: 17
##   GLL at iteration 17 : -4967.144
## Iteration: 18
##   GLL at iteration 18 : -4868.046
## Iteration: 19
##   GLL at iteration 19 : -4982.152
## Iteration: 20
##   GLL at iteration 20 : -4821.064
## Iteration: 21
##   GLL at iteration 21 : -5000.843
## Iteration: 22
##   GLL at iteration 22 : -5007.34
```

```
## Iteration: 23
##   GLL at iteration 23 : -4990.698
## Iteration: 24
##   GLL at iteration 24 : -4944.376
## Iteration: 25
##   GLL at iteration 25 : -4996.544
## Iteration: 26
##   GLL at iteration 26 : -5042.552
## Iteration: 27
##   GLL at iteration 27 : -5041.11
## Iteration: 28
##   GLL at iteration 28 : -4985.112
## Iteration: 29
##   GLL at iteration 29 : -4996.289
## Iteration: 30
##   GLL at iteration 30 : -4940.108
## Iteration: 31
##   GLL at iteration 31 : -5027.016
## Iteration: 32
##   GLL at iteration 32 : -5167.092
## Iteration: 33
##   GLL at iteration 33 : -4954.408
## Iteration: 34
##   GLL at iteration 34 : -5061.333
## Iteration: 35
##   GLL at iteration 35 : -5093.463
## Iteration: 36
##   GLL at iteration 36 : -5089.323
## Iteration: 37
##   GLL at iteration 37 : -5048.276
## Iteration: 38
##   GLL at iteration 38 : -5101.582
## Iteration: 39
##   GLL at iteration 39 : -5049.8
## Iteration: 40
##   GLL at iteration 40 : -5004.022
## Iteration: 41
##   GLL at iteration 41 : -5080.773
## Iteration: 42
##   GLL at iteration 42 : -5041.33
## Iteration: 43
##   GLL at iteration 43 : -5094.423
## Iteration: 44
##   GLL at iteration 44 : -5148.542
## Iteration: 45
##   GLL at iteration 45 : -5090.214
## Iteration: 46
##   GLL at iteration 46 : -4994.566
## Iteration: 47
##   GLL at iteration 47 : -5025.193
## Iteration: 48
##   GLL at iteration 48 : -5133.169
## Iteration: 49
##   GLL at iteration 49 : -5150.384
```

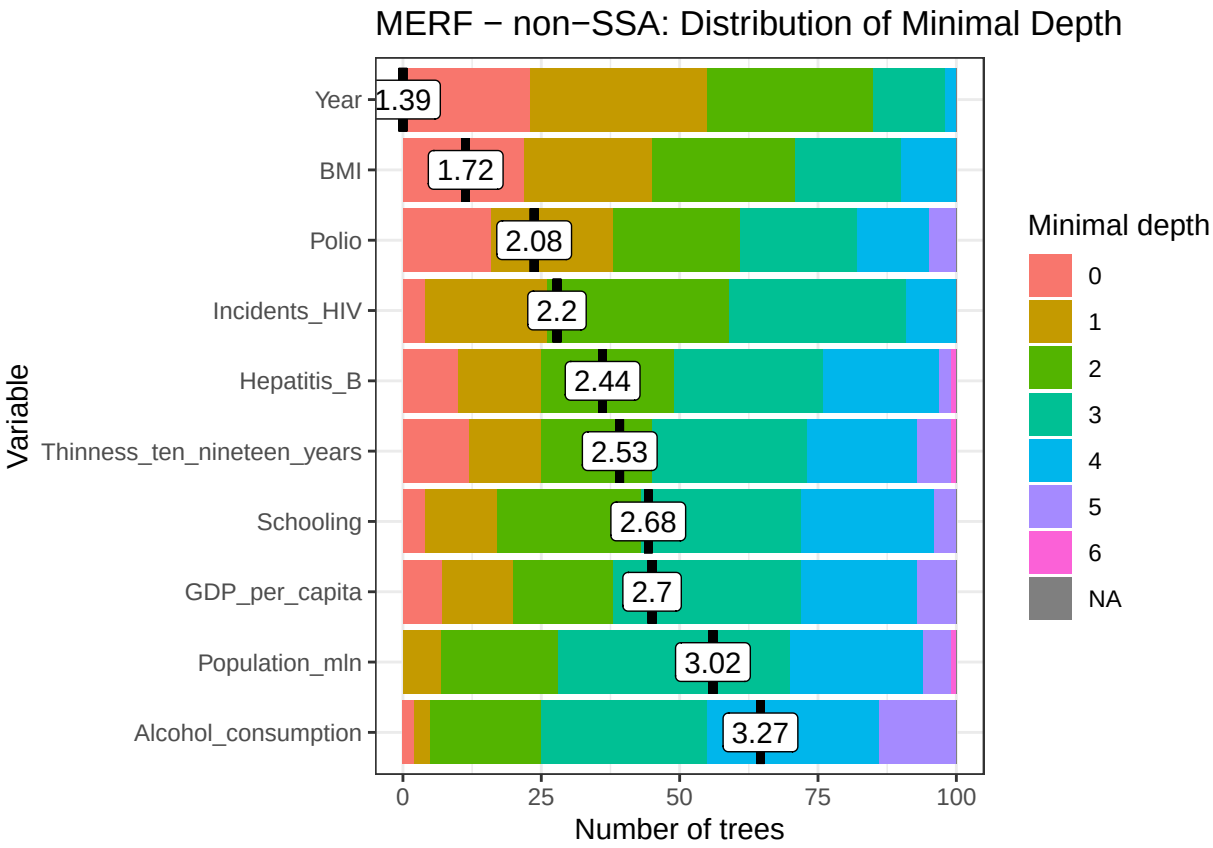
```
## Iteration: 50
##   GLL at iteration 50 : -5251.648
## Iteration: 51
##   GLL at iteration 51 : -5134.279
## Iteration: 52
##   GLL at iteration 52 : -5068.066
## Iteration: 53
##   GLL at iteration 53 : -5171.11
## Iteration: 54
##   GLL at iteration 54 : -5093.865
## Iteration: 55
##   GLL at iteration 55 : -5203.212
## Iteration: 56
##   GLL at iteration 56 : -5055.788
## Iteration: 57
##   GLL at iteration 57 : -5109.987
## Iteration: 58
##   GLL at iteration 58 : -5054.047
## Iteration: 59
##   GLL at iteration 59 : -5146.066
## Iteration: 60
##   GLL at iteration 60 : -5079.741
## Iteration: 61
##   GLL at iteration 61 : -5216.878
## Iteration: 62
##   GLL at iteration 62 : -5129.881
## Iteration: 63
##   GLL at iteration 63 : -5038.226
## Iteration: 64
##   GLL at iteration 64 : -5162.798
## Iteration: 65
##   GLL at iteration 65 : -5162.09
## Iteration: 66
##   GLL at iteration 66 : -5159.78
## Iteration: 67
##   GLL at iteration 67 : -5190.816
## Iteration: 68
##   GLL at iteration 68 : -5207.431
## Iteration: 69
##   GLL at iteration 69 : -5150.782
## Iteration: 70
##   GLL at iteration 70 : -5123.172
## Iteration: 71
##   GLL at iteration 71 : -5235.401
## Iteration: 72
##   GLL at iteration 72 : -5255.051
## Iteration: 73
##   GLL at iteration 73 : -5266.83
## Iteration: 74
##   GLL at iteration 74 : -5279.094
## Iteration: 75
##   GLL at iteration 75 : -5173.012
## Iteration: 76
##   GLL at iteration 76 : -5215.246
```

```
## Iteration: 77
##   GLL at iteration 77 : -5127.995
## Iteration: 78
##   GLL at iteration 78 : -5246.531
## Iteration: 79
##   GLL at iteration 79 : -5139.302
## Iteration: 80
##   GLL at iteration 80 : -5204.206
## Iteration: 81
##   GLL at iteration 81 : -5210.995
## Iteration: 82
##   GLL at iteration 82 : -5162.811
## Iteration: 83
##   GLL at iteration 83 : -5265.911
## Iteration: 84
##   GLL at iteration 84 : -5202.364
## Iteration: 85
##   GLL at iteration 85 : -5360.429
## Iteration: 86
##   GLL at iteration 86 : -5176.659
## Iteration: 87
##   GLL at iteration 87 : -5264.306
## Iteration: 88
##   GLL at iteration 88 : -5259.112
## Iteration: 89
##   GLL at iteration 89 : -5301.981
## Iteration: 90
##   GLL at iteration 90 : -5241.345
## Iteration: 91
##   GLL at iteration 91 : -5218.222
## Iteration: 92
##   GLL at iteration 92 : -5259.467
## Iteration: 93
##   GLL at iteration 93 : -5270.914
## Iteration: 94
##   GLL at iteration 94 : -5201.431
## Iteration: 95
##   GLL at iteration 95 : -5285.889
## Iteration: 96
##   GLL at iteration 96 : -5215.464
## Iteration: 97
##   GLL at iteration 97 : -5327.847
## Iteration: 98
##   GLL at iteration 98 : -5165.359
## Iteration: 99
##   GLL at iteration 99 : -5317.051
## Iteration: 100
##   GLL at iteration 100 : -5182.379
## Iteration: 101
##   GLL at iteration 101 : -5223.988
## Iteration: 102
##   GLL at iteration 102 : -5256.744
## Iteration: 103
##   GLL at iteration 103 : -5174.496
```

```
## Iteration: 104
##   GLL at iteration 104 : -5208.92
## Iteration: 105
##   GLL at iteration 105 : -5258.468
## Iteration: 106
##   GLL at iteration 106 : -5200.694
## Iteration: 107
##   GLL at iteration 107 : -5252.261
## Iteration: 108
##   GLL at iteration 108 : -5171.96
## Iteration: 109
##   GLL at iteration 109 : -5255.649
## Iteration: 110
##   GLL at iteration 110 : -5231.211
## Iteration: 111
##   GLL at iteration 111 : -5255.612
## Iteration: 112
##   GLL at iteration 112 : -5131.329
## Iteration: 113
##   GLL at iteration 113 : -5247.911
## Iteration: 114
##   GLL at iteration 114 : -5248.81
## Iteration: 115
##   GLL at iteration 115 : -5244.364
## Iteration: 116
##   GLL at iteration 116 : -5255.823
## Iteration: 117
##   GLL at iteration 117 : -5339.373
## Iteration: 118
##   GLL at iteration 118 : -5202.109
## Iteration: 119
##   GLL at iteration 119 : -5305.922
## Iteration: 120
##   GLL at iteration 120 : -5295.001
## Iteration: 121
##   GLL at iteration 121 : -5125.097
## Iteration: 122
##   GLL at iteration 122 : -5340.519
## Iteration: 123
##   GLL at iteration 123 : -5246.503
## Iteration: 124
##   GLL at iteration 124 : -5345.647
## Iteration: 125
##   GLL at iteration 125 : -5309.982
## Iteration: 126
##   GLL at iteration 126 : -5259.616
## Iteration: 127
##   GLL at iteration 127 : -5318.159
## Iteration: 128
##   GLL at iteration 128 : -5327.169
## Iteration: 129
##   GLL at iteration 129 : -5383.847
## Iteration: 130
##   GLL at iteration 130 : -5278.665
```

```
## Iteration: 131
##   GLL at iteration 131 : -5378.548
## Iteration: 132
##   GLL at iteration 132 : -5215.689
## Iteration: 133
##   GLL at iteration 133 : -5259.442
## Iteration: 134
##   GLL at iteration 134 : -5276.168
## Iteration: 135
##   GLL at iteration 135 : -5273.663
## Iteration: 136
##   GLL at iteration 136 : -5363.517
## Iteration: 137
##   GLL at iteration 137 : -5316.052
## Iteration: 138
##   GLL at iteration 138 : -5326.868
## Iteration: 139
##   GLL at iteration 139 : -5138.309
## Iteration: 140
##   GLL at iteration 140 : -5386.6
## Iteration: 141
##   GLL at iteration 141 : -5249.739
## Iteration: 142
##   GLL at iteration 142 : -5303.394
## Iteration: 143
##   GLL at iteration 143 : -5119.012
## Iteration: 144
##   GLL at iteration 144 : -5270.805
## Iteration: 145
##   GLL at iteration 145 : -5288.687
## Iteration: 146
##   GLL at iteration 146 : -5305.183
## Iteration: 147
##   GLL at iteration 147 : -5240.014
## Iteration: 148
##   GLL at iteration 148 : -5122.685
## Iteration: 149
##   GLL at iteration 149 : -5327.565
## Iteration: 150
##   GLL at iteration 150 : -5320.503
```

```r
# non SSA
min_depth_non_ssa <- min_depth_distribution(merf_result_non_SSA$rf_model)
plot_min_depth_distribution(min_depth_ssa, mean_sample = "all_trees") +
  ggtitle("MERF - non-SSA: Distribution of Minimal Depth")
```

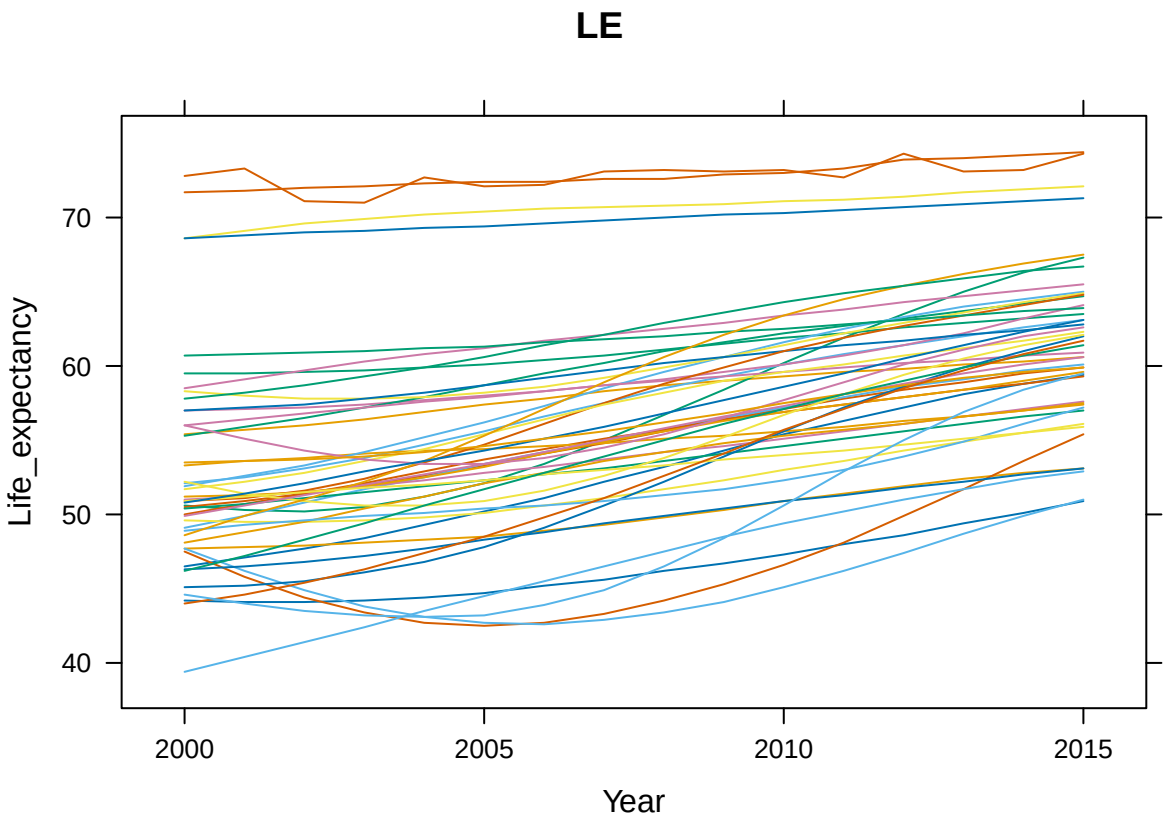


The LME for non-SSA region significantly better - closer to MERF.

```
length(unique(SSA$Country))
```

```
## [1] 47
```

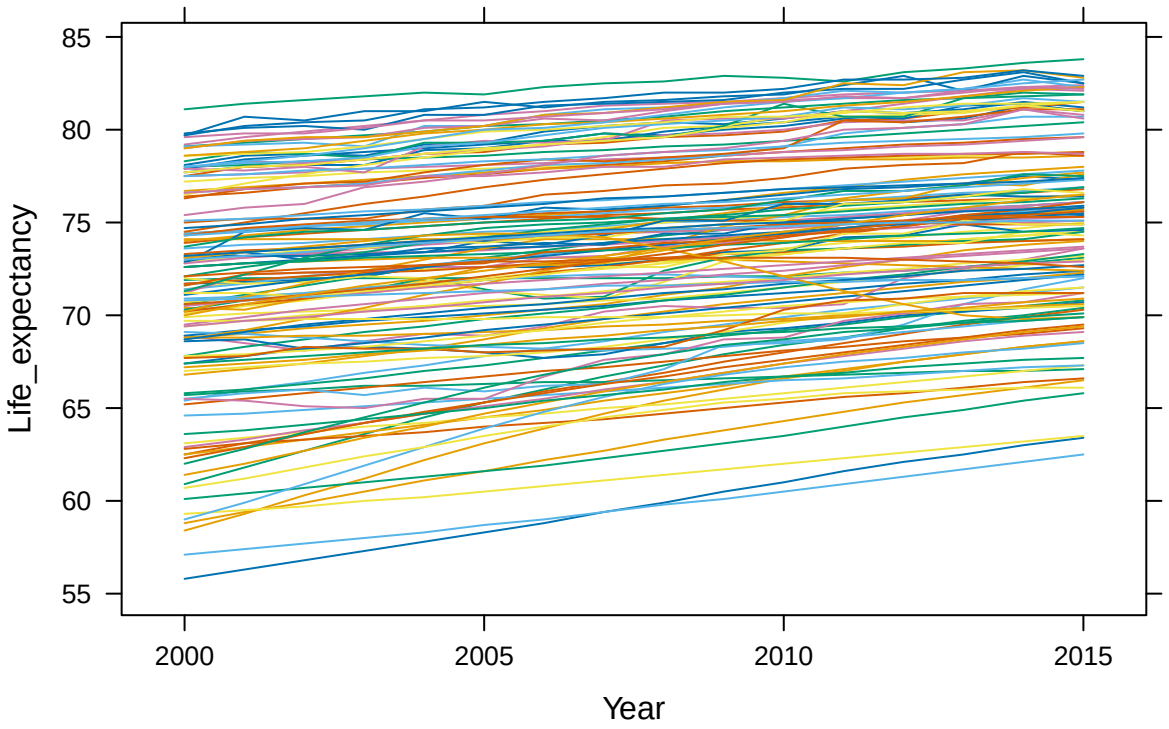
```
xyplot(Life_expectancy ~ Year, groups = Country, data = SSA[order(SSA$Country, SSA$Year),], type = "l",
```




```
length(unique(non_SSA$Country))
```

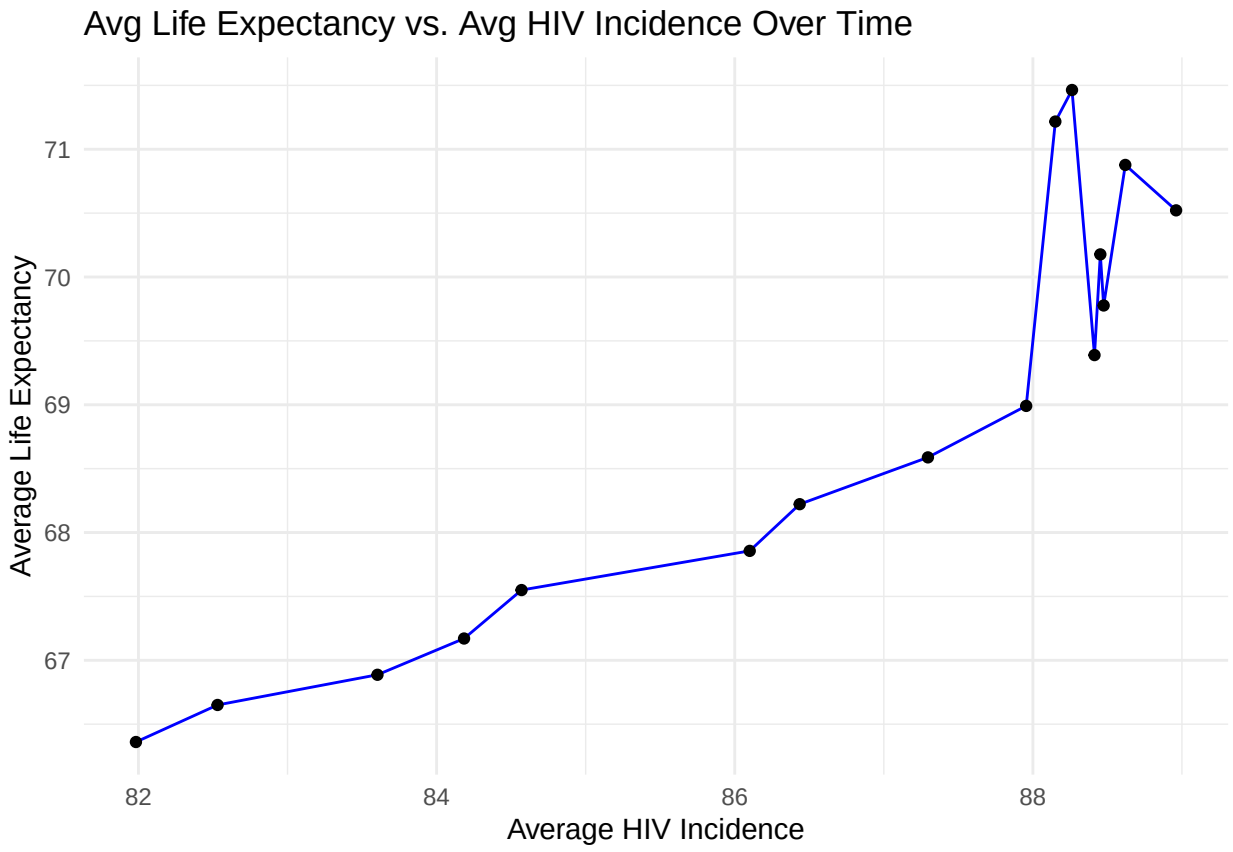
```
## [1] 132
```

```
xyplot(Life_expectancy ~ Year, groups = Country, data = non_SSA[order(non_SSA$Country, non_SSA$Year),],
```

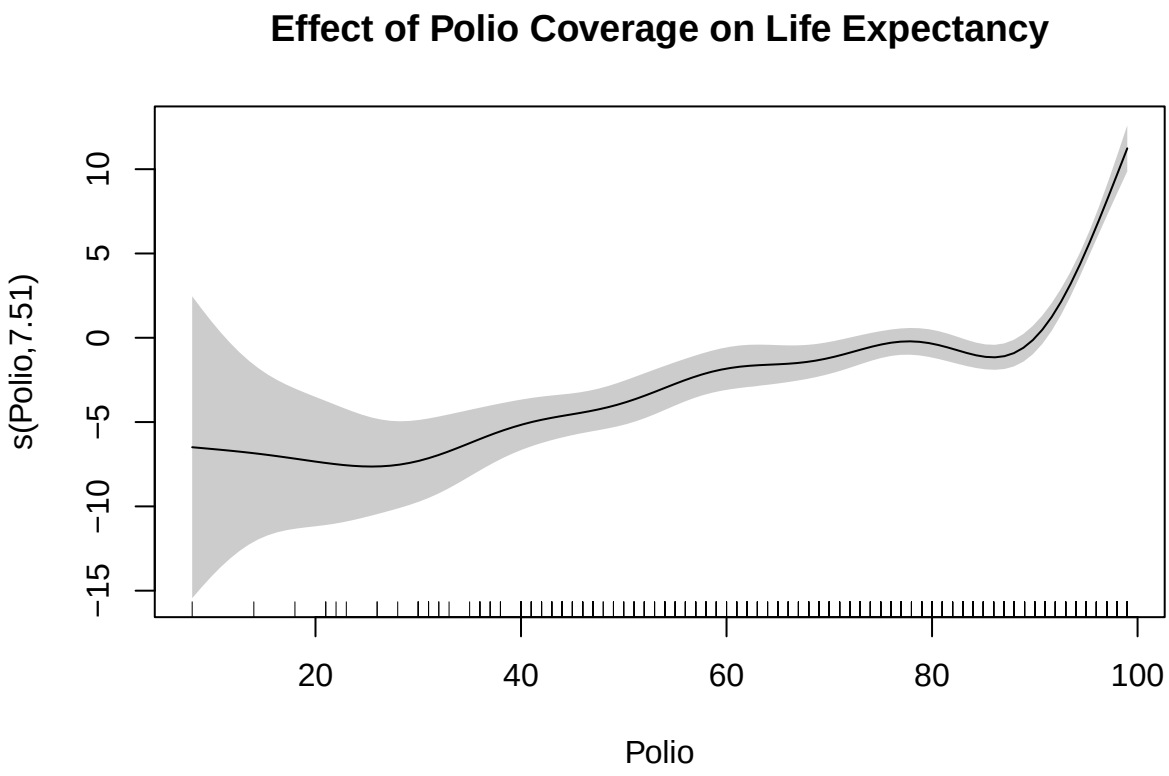



```
life_summary <- life_final %>%
  group_by(Year) %>%
  summarise(
    avg_hiv = mean(Polio, na.rm = TRUE),
    avg_life = mean(Life_expectancy, na.rm = TRUE)
  )

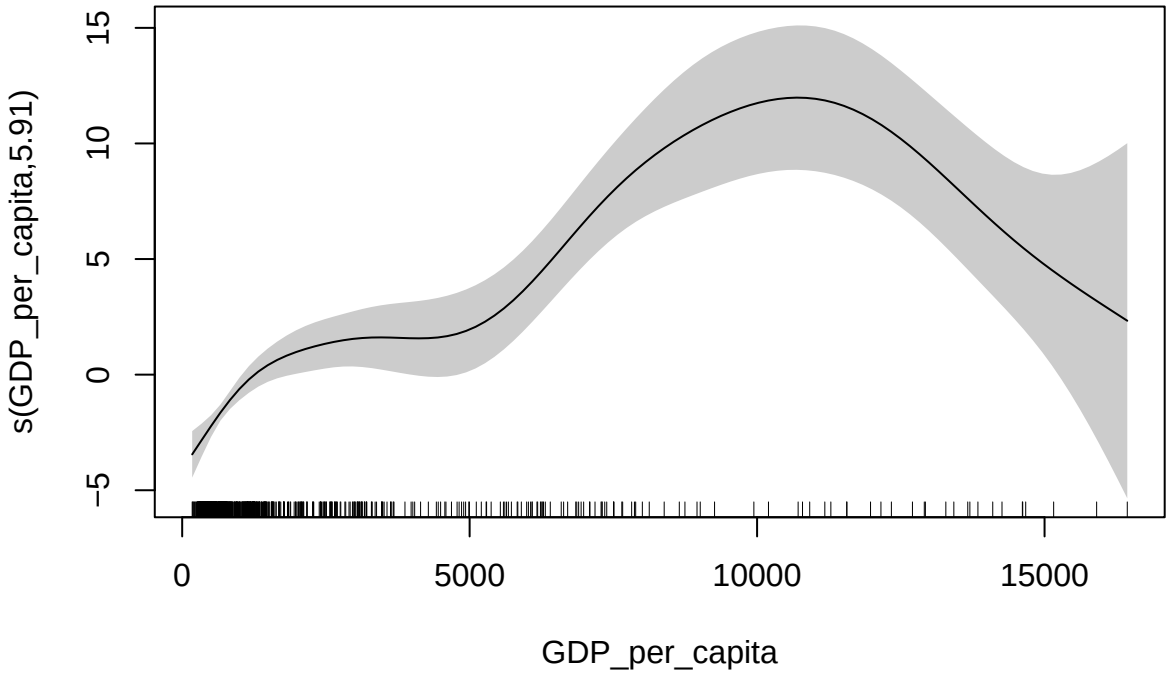
ggplot(life_summary, aes(x = avg_hiv, y = avg_life)) +
  geom_line(color = "blue") +
  geom_point() +
  labs(title = "Avg Life Expectancy vs. Avg HIV Incidence Over Time",
       x = "Average HIV Incidence",
       y = "Average Life Expectancy") +
  theme_minimal()
```



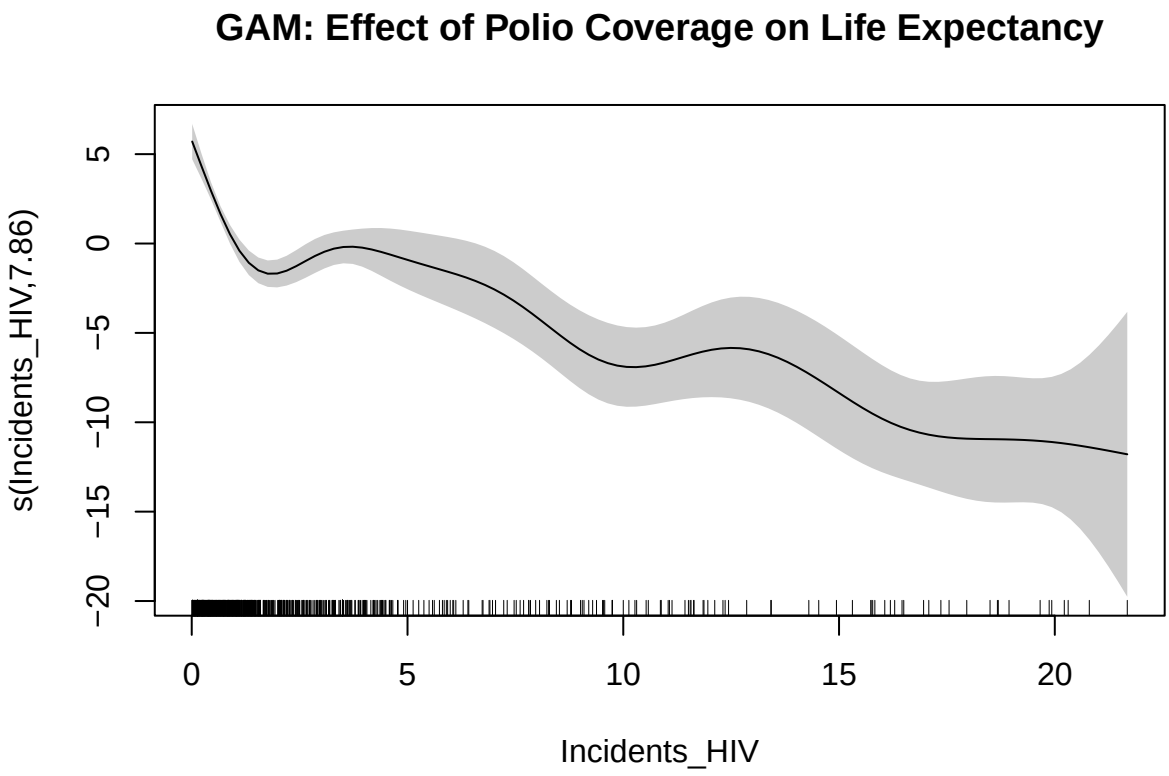

```r
library(mgcv)
gam_model <- gam(Life_expectancy ~ s(Polio), data = SSA)
plot(gam_model, shade = TRUE, main = "Effect of Polio Coverage on Life Expectancy")
```



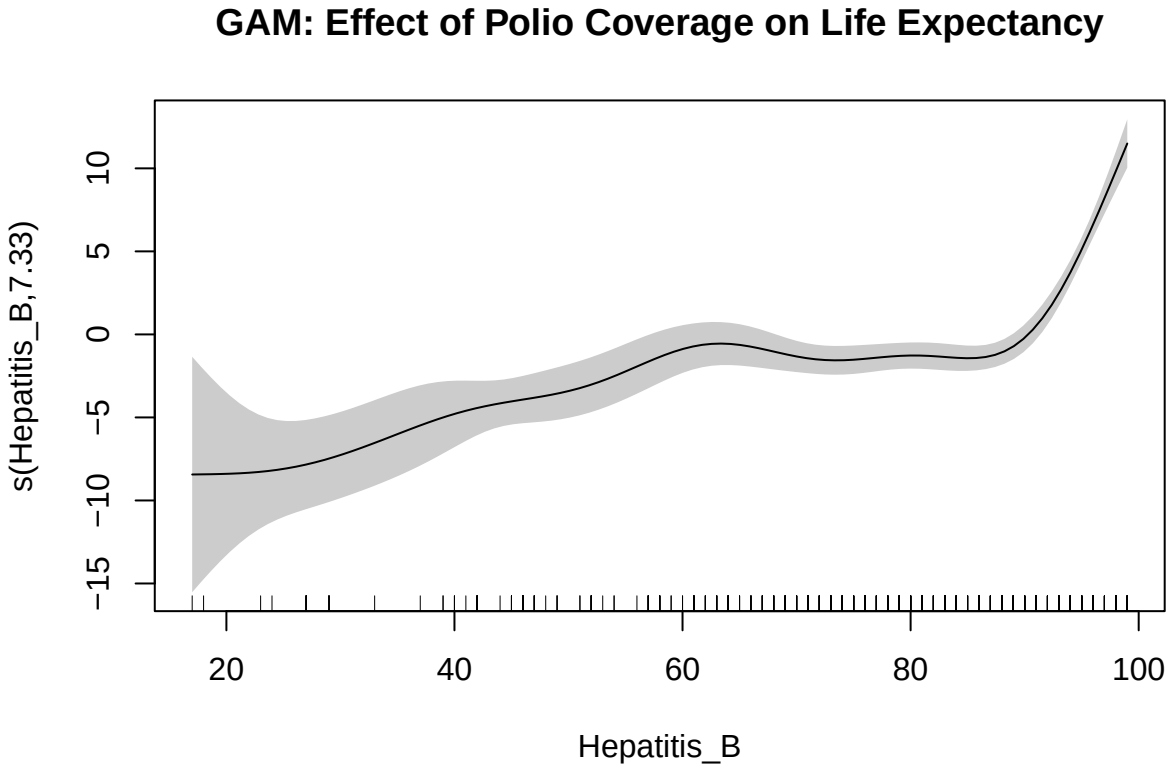

```r
library(mgcv)
gam_model <- gam(Life_expectancy ~ s(GDP_per_capita), data = SSA)
plot(gam_model, shade = TRUE, main = "Effect of GDP on Life Expectancy")
```



```r
library(mgcv)
gam_model <- gam(Life_expectancy ~ s(Incidents_HIV), data = SSA)
plot(gam_model, shade = TRUE, main = "GAM: Effect of Polio Coverage on Life Expectancy")
```




```r
library(mgcv)
gam_model <- gam(Life_expectancy ~ s(Hepatitis_B), data = SSA)
plot(gam_model, shade = TRUE, main = "GAM: Effect of Polio Coverage on Life Expectancy")
```


GAM: Effect of Polio Coverage on Life Expectancy
s(Hepatitis_B,7.33)
10
5
0
−5
−10
−15
20
40
60
80
100
Hepatitis_B